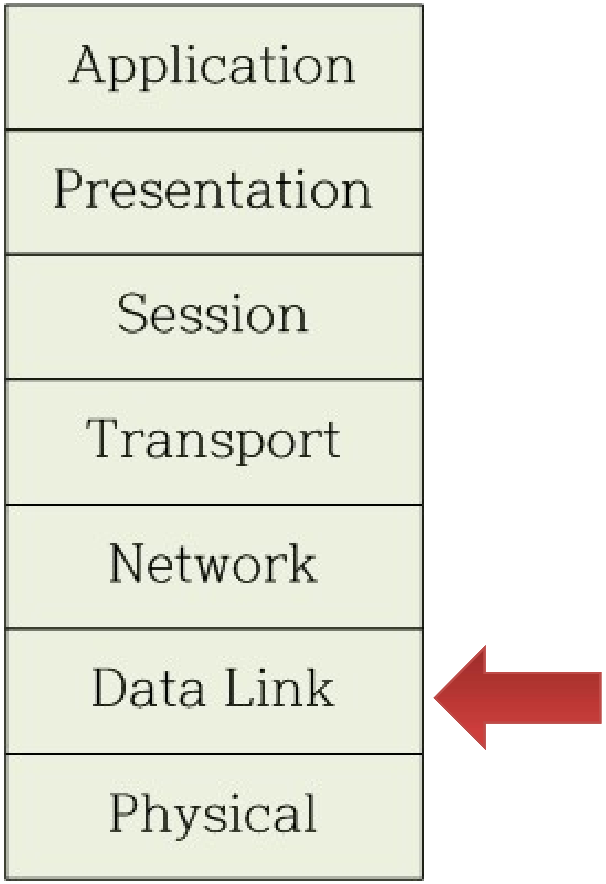
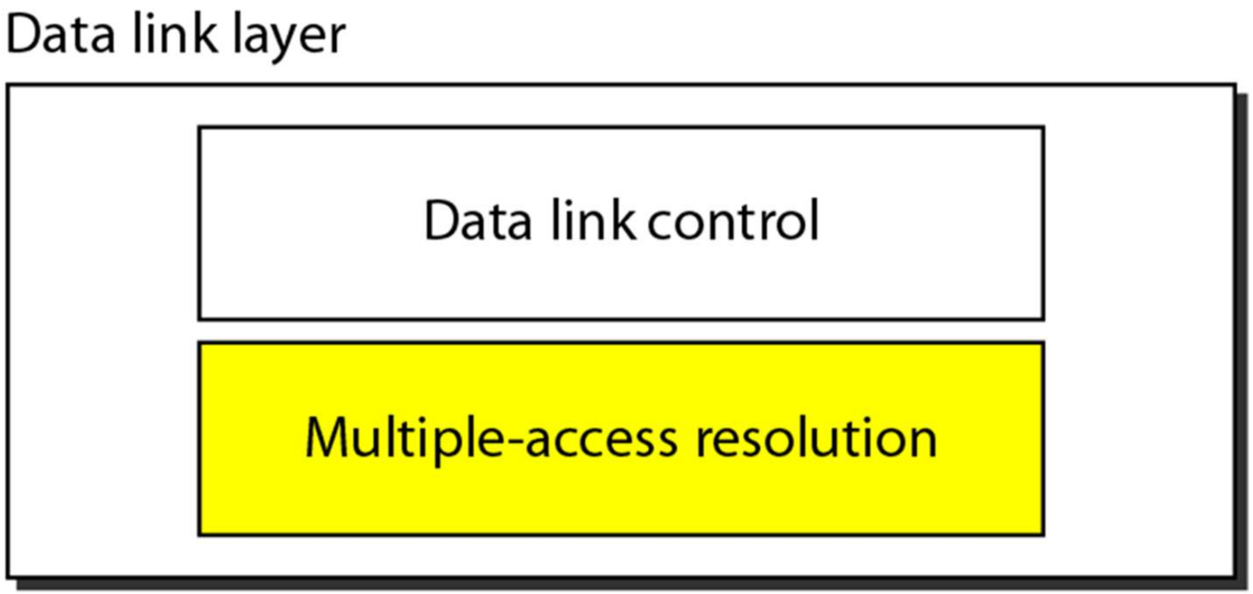
Medium Access Control

Now we are here

- OSI 7 layer model

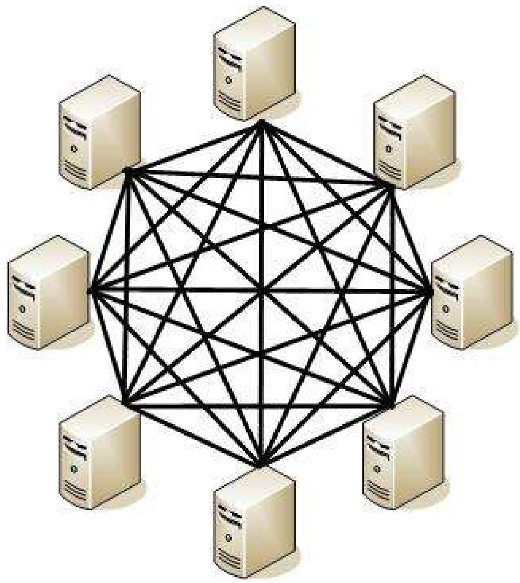


Components of data link layer



What is multiple access?

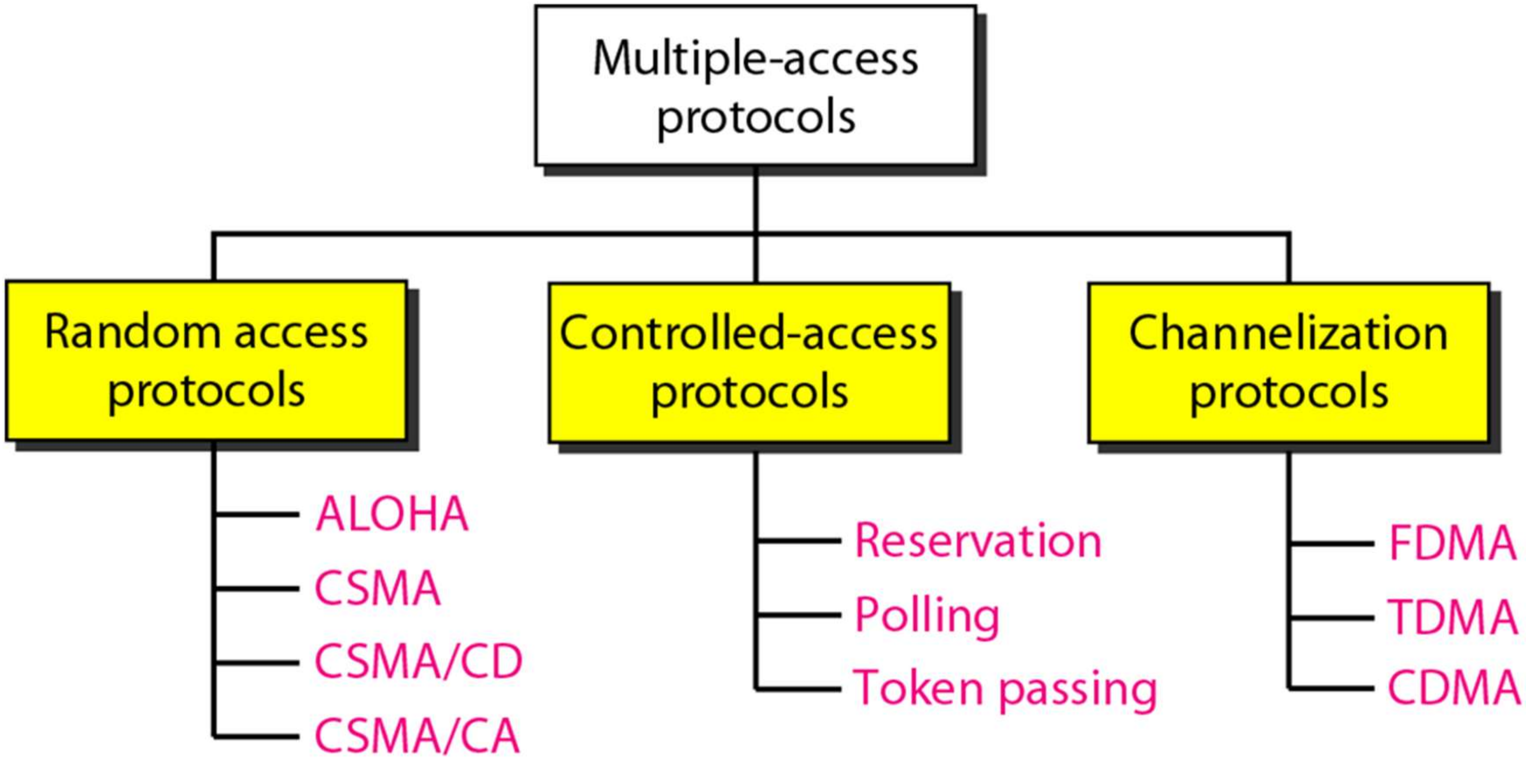
- Techniques to share a common link
- point-to-point vs. shared
- In shared link, only one user should transmit at a time → medium access control necessary



point-to-point

shared

Types of multiple access protocols



Random Access Protocols

Random access protocols

- Collisions are avoided using "randomness"
- Similar to a group of people chatting in a room
  - No one knows when a person will start talking
  - If two person starts to talk at the same time, they both stop, wait for a moment, and start to talk again

ALOHA

- Developed in 1970s at University of Hawaii
  - Designed for wireless communication between the Hawaiian islands

- A basic protocol in the line of random access protocols

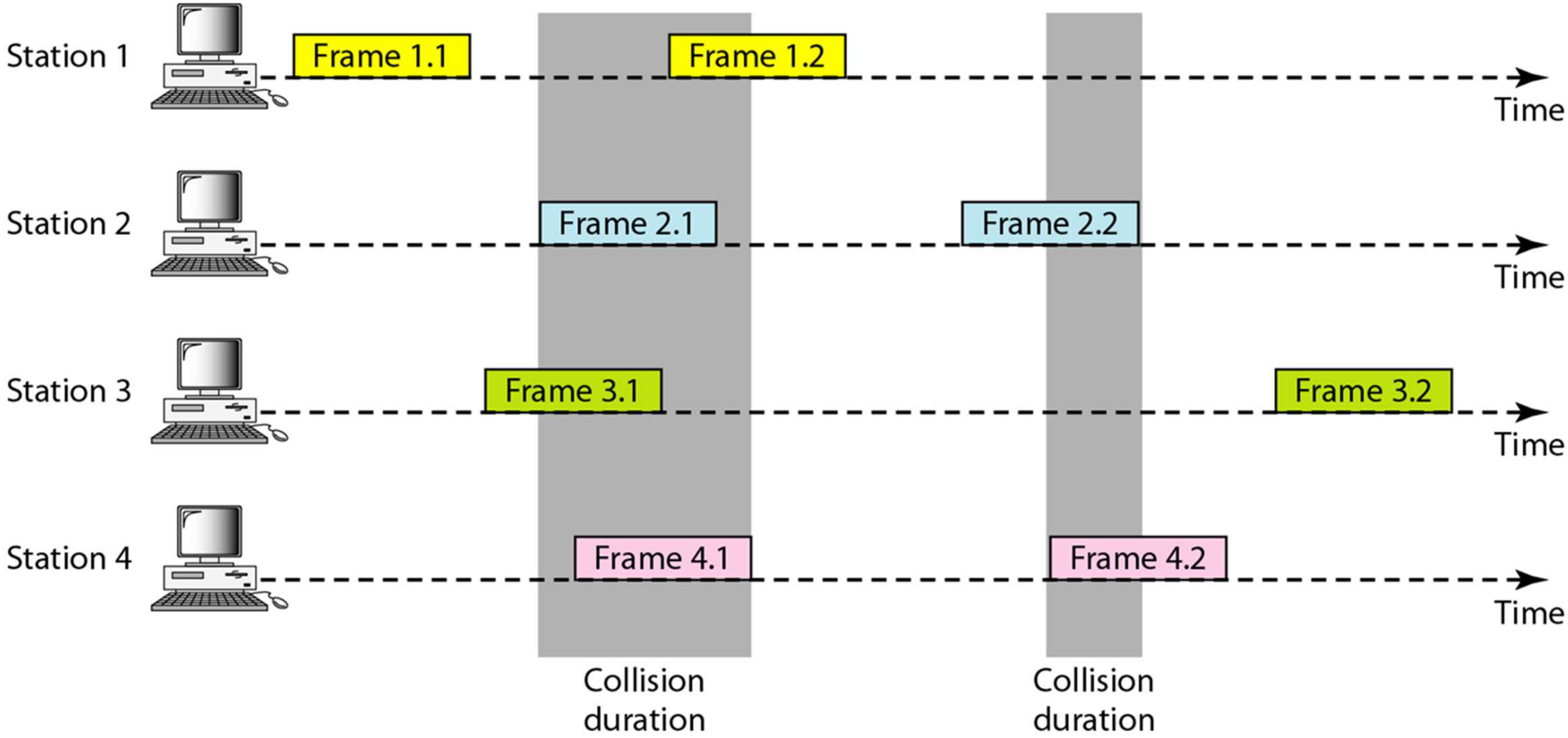
- Summary
  - If there is anything to send, send it.
  - If collision occurs, wait for a certain duration of time and retransmit

Pure ALOHA

- If sender has a frame to send, send it immediately.
- When receiver receives the frame, send ACK.
- If ACK does not arrive during timeout period, sender retransmits the frame
- However, the sender does not retransmit immediately; <u>the sender waits for a random delay and retransmits</u>

Pure ALOHA

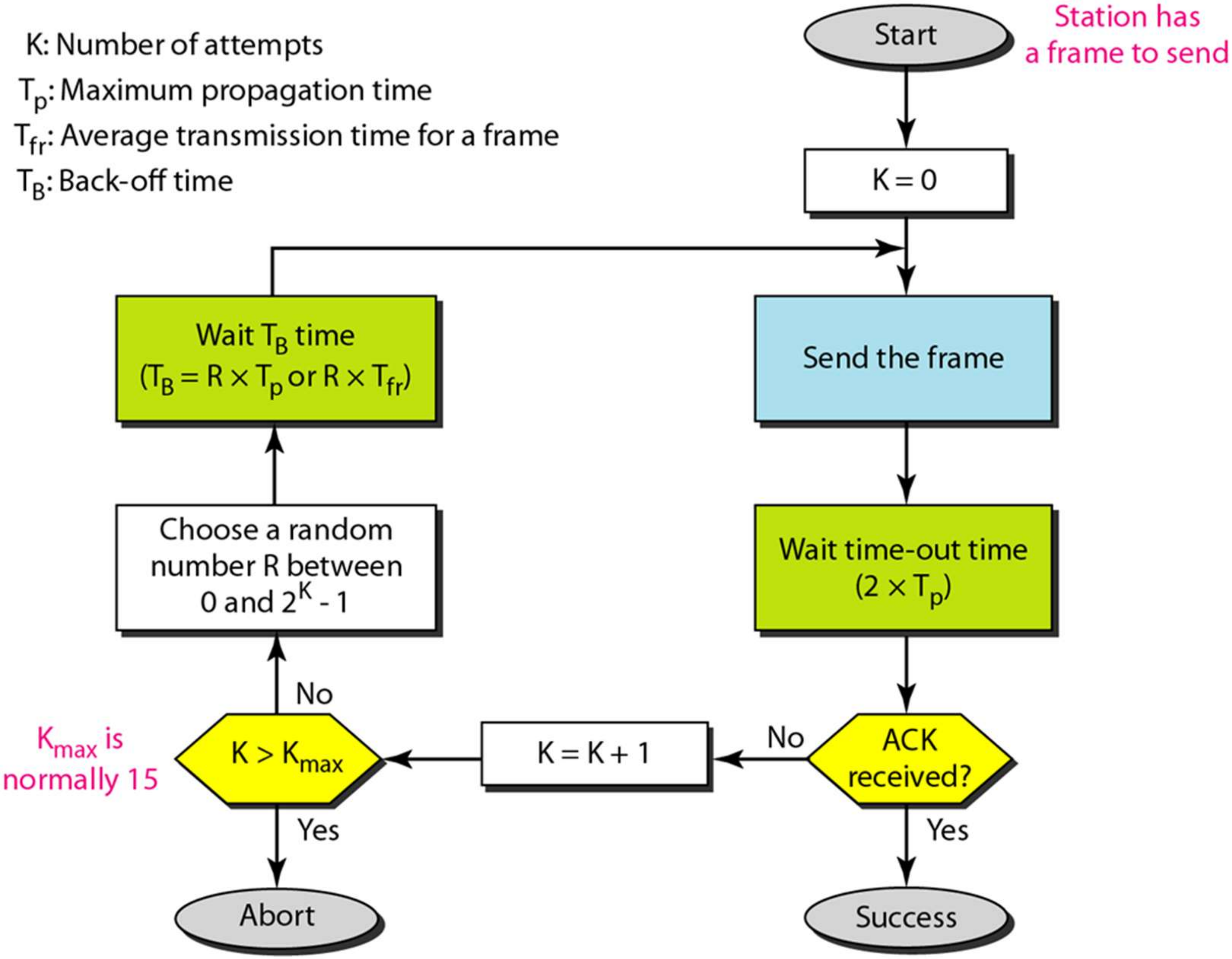
- If two frames are on the link at the same time, a collision occurs



Pure ALOHA

- What does the sender do when a frame is lost?
  - Wait for some random time and retransmit
- How long should it wait?
  - Choose a random number R between 0 and 2^K-1
  - Wait for T x R seconds (T: time slot)
  - K is incremented after each timeout
    → binary exponential backoff

Pure ALOHA



K: Number of attempts
T_p: Maximum propagation time
T_{fr}: Average transmission time for a frame
T_B: Back-off time
Start
Station has a frame to send
K = 0
Send the frame
Wait time-out time ($2 \times T_p$)
ACK received?
No
Yes
Success
K = K + 1
$K > K_{max}$
K_{max} is normally 15
No
Yes
Abort
Choose a random number R between 0 and $2^K - 1$
Wait T_B time ($T_B = R \times T_p$ or $R \times T_{fr}$)

Pure ALOHA

- Time slot T
  - Fixed (T_s)
  - Transmission time (T_{fr}): when T_{fr} >> T_p
  - Maximum propagation time (T_p): when T_p >> T_{fr}

- In practice, often speficied in the standard
  - e.g., Ethernet: 512 bit times (51.2μs at 10Mbps)
  - e.g., Wi-Fi: 9μs (802.11a/g/n/ac)

Example

- A network of nodes uses pure ALOHA as medium access protocol

- Backoff time T_B is calculated as follows:
  - $T_B = T \times R$
  - T (time slot) = 20ms
  - $R = rand(0, 2^K-1)$

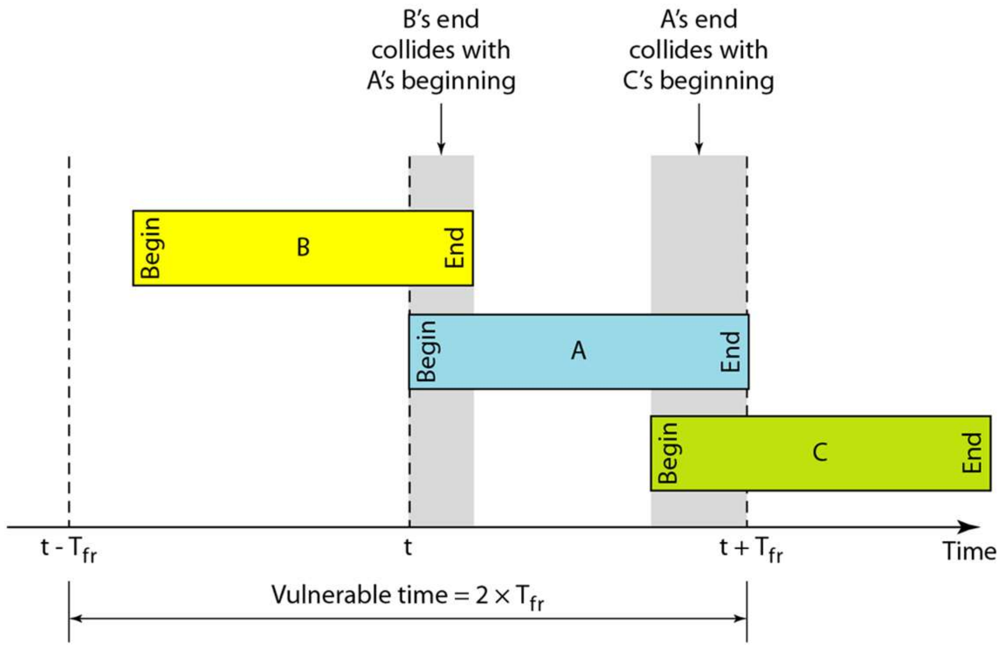
- If K = 1, what are the possible backoff durations?

Vulnerable time

- Suppose a frame is transmitted on the channel. The vulnerable time is the duration of time in which a collision occurs, if another frame is transmitted.

Vulnerable time

- Vulnerable time of pure ALOHA
  - Assume T_p is minimal compared to T_{fr}
  - Suppose A's transmission starts at t and finishes at $t + T_{fr}$
  - No one should transmit between $t – T_{fr}$ and $t + T_{fr}$
  - Vulnerable time: $2 \times T_{fr}$



Example

- A pure ALOHA network transmits 200-bit frames on a shared channel of 200kbps. What is the requirement to make this frame collision-free? (neglect propagation delay)

Throughput

- To calculate throughput of pure ALOHA, we compute S as follows

$$S = G \times e^{-2G}$$

- S: ratio of frames that are successfully transmitted
- G: average number of frames generated in the system during T_{fr}

Throughput

- If G = 1,
- $S = 1 \times e^{-2} = 0.135$
- If 1000 frames are generated per second in the system, 135 frames can be successfully sent
- $e^{-1} = 0.367$
- $e^{-2} = 0.135$
- $e^{-0.5} = 0.606$

Example

- A pure ALOHA network transmits 200-bit frames on a shared channel of 200kbps. What is the number of frames that are successfully sent if the system (all stations together) produces

- a. 1000 frames per second
- b. 500 frames per second
- c. 250 frames per second

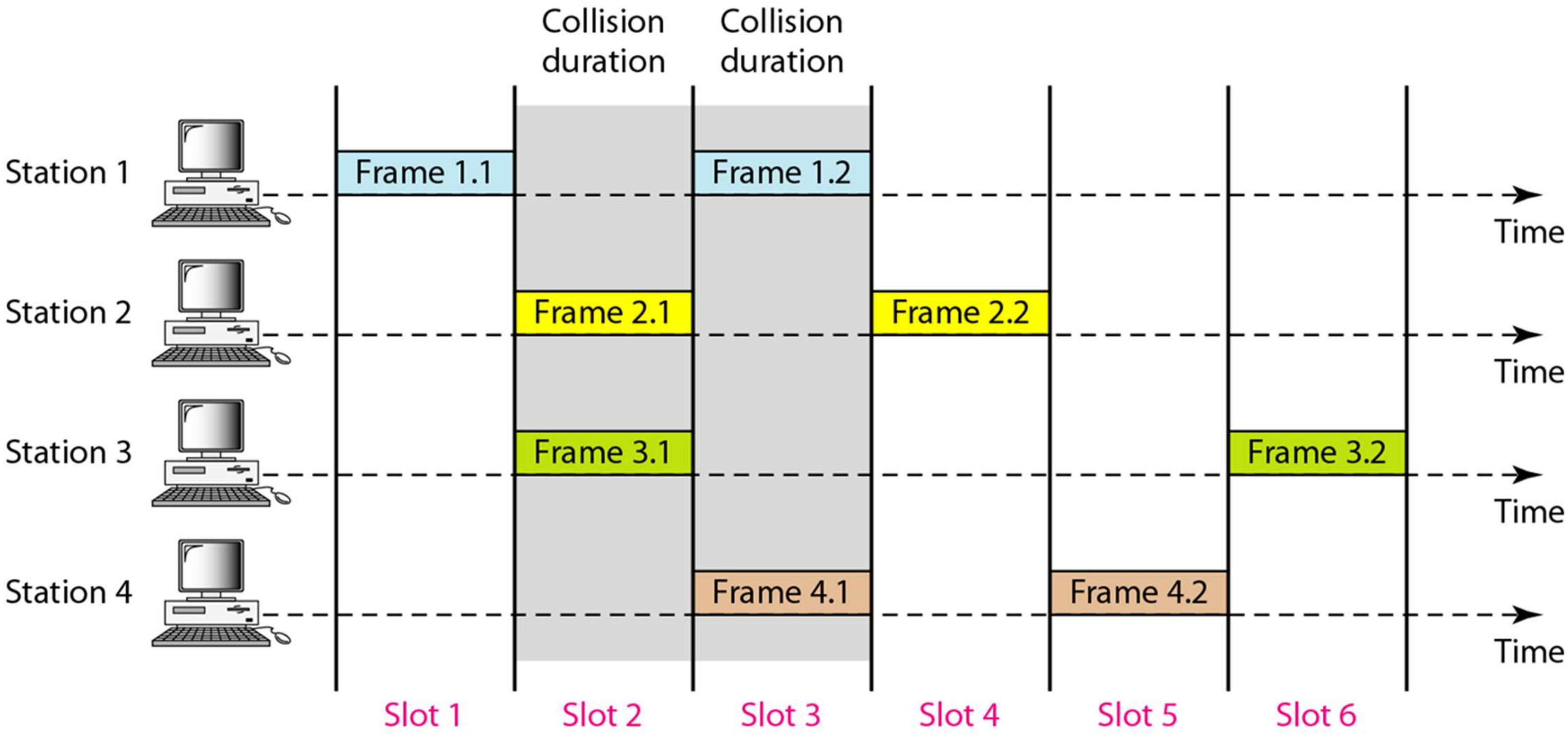
Solution for b.

- T_{fr} = 200 bits / 200 kbps = 1ms
- If the system creates 500 frames per second, it is 0.5 frames per T_{fr} → G = 0.5
- $S = G \times e^{-2G} = 0.184$
- So in this case, 184 frames can be successfully sent (and received) per second.

Slotted ALOHA

- Problem of Pure ALOHA
  - Maximum throughput is only about 18.4%
  - Because vulnerable time is $2 \cdot T_{fr}$
- Idea of Slotted ALOHA
  - Divide time into slots (each slot = T_{fr})
  - Nodes can only transmit at the beginning of a slot
  - i.e., "can send anytime" → "can only send at specific times"
- Effect
  - Vulnerable time is reduced to T_{fr}
  - Maximum throughput doubles to about 36.8%
  - Cost: all nodes need time synchronization

Slotted ALOHA



Slotted ALOHA

- Throughput of slotted ALOHA
- $S = G \times e^{-G}$
- Suppose we send 200-bit frames at 200kbps. How many frames can be successfully sent if
  - 1000 frames are generated per second
  - 500 frames are generated per second
  - 250 frames are generated per second

CSMA (Carrier Sense Multiple Access)

- In ALOHA (pure/slotted), a node starts transmission even if other nodes are transmitting.
  - If the node is lucky, no other node transmits in the vulnerable time period
  - If the node is unlucky, a collision occurs

- If we use hardware that can sense the medium (channel), we can further reduce collision

- CSMA: "Listen before you talk"

CSMA

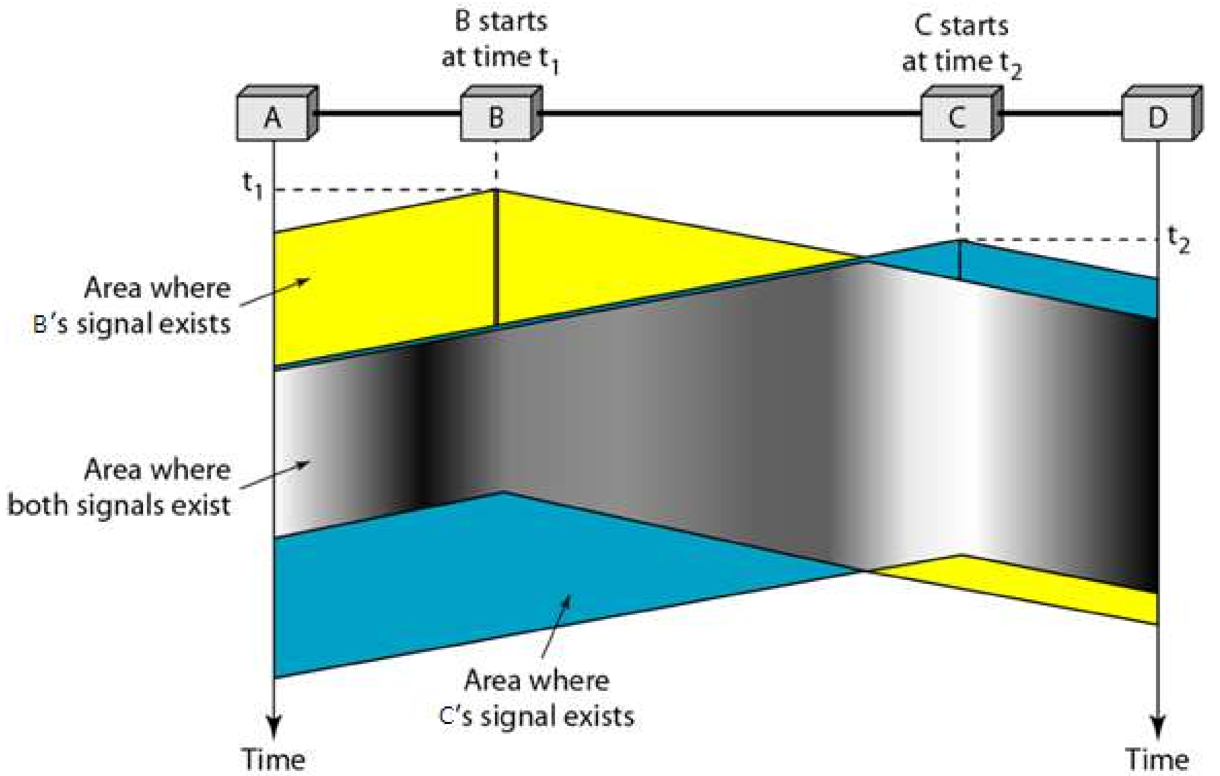
- When a node has a frame to transmit
- It first senses the channel
- If there is no signal on the channel (<u>idle</u>), go ahead and transmit
- If there is signal on the channel (<u>busy</u>), do not transmit

CSMA

- Human analogy
  - Suppose a group of people are chatting in a room
  - I want to talk
  - If no one is talking, I start to talk
  - If someone is talking, I wait

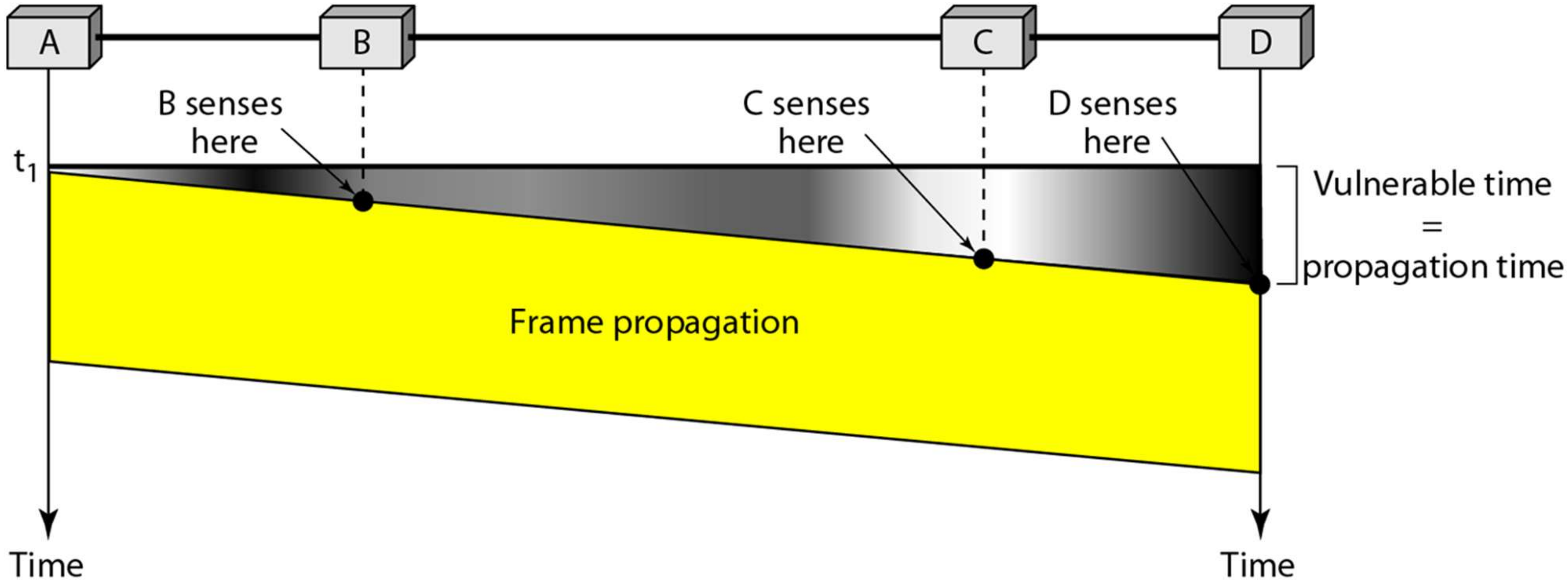
CSMA

- Can CSMA avoid collision completely? No.
  - If B transmits at time t_1, it reaches C at t_3
  - At t_2 ($t_1 < t_2 < t_3$), C sees channel as idle (no signal).
  - If C transmits at t_2 a collision occurs.



CSMA

- Vulnerable time: T_p (propagation delay)



CSMA

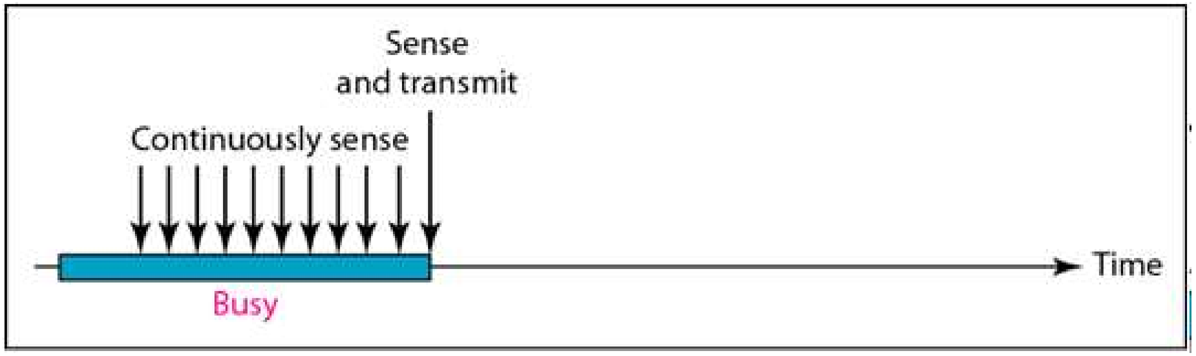
- If the channel is busy, what should the node do?

- Depending on the action, there are 3 types of CSMA.

- 1-persistent CSMA
- non-persistent CSMA
- p-persistent CSMA

1-persistent CSMA

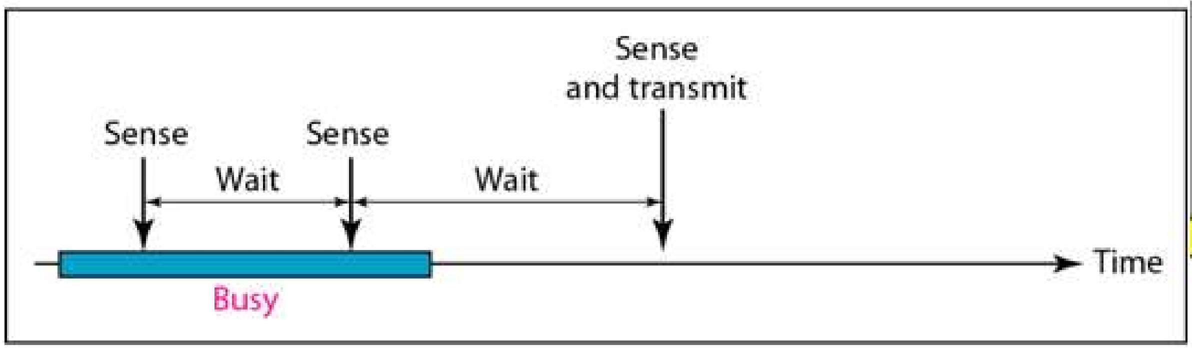
- Continuously sense the channel while the channel is busy.
- As soon as the channel becomes idle, transmit



- Since multiple nodes can wait for the channel to become idle, collision probability is high.

Non-persistent CSMA

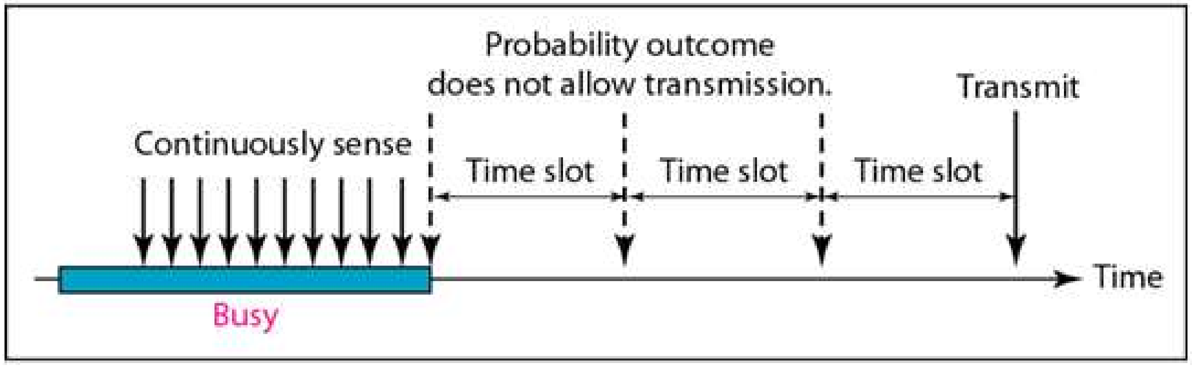
- If the channel is busy, the node waits for some time and sense the channel again.
- If the channel is idle, transmit



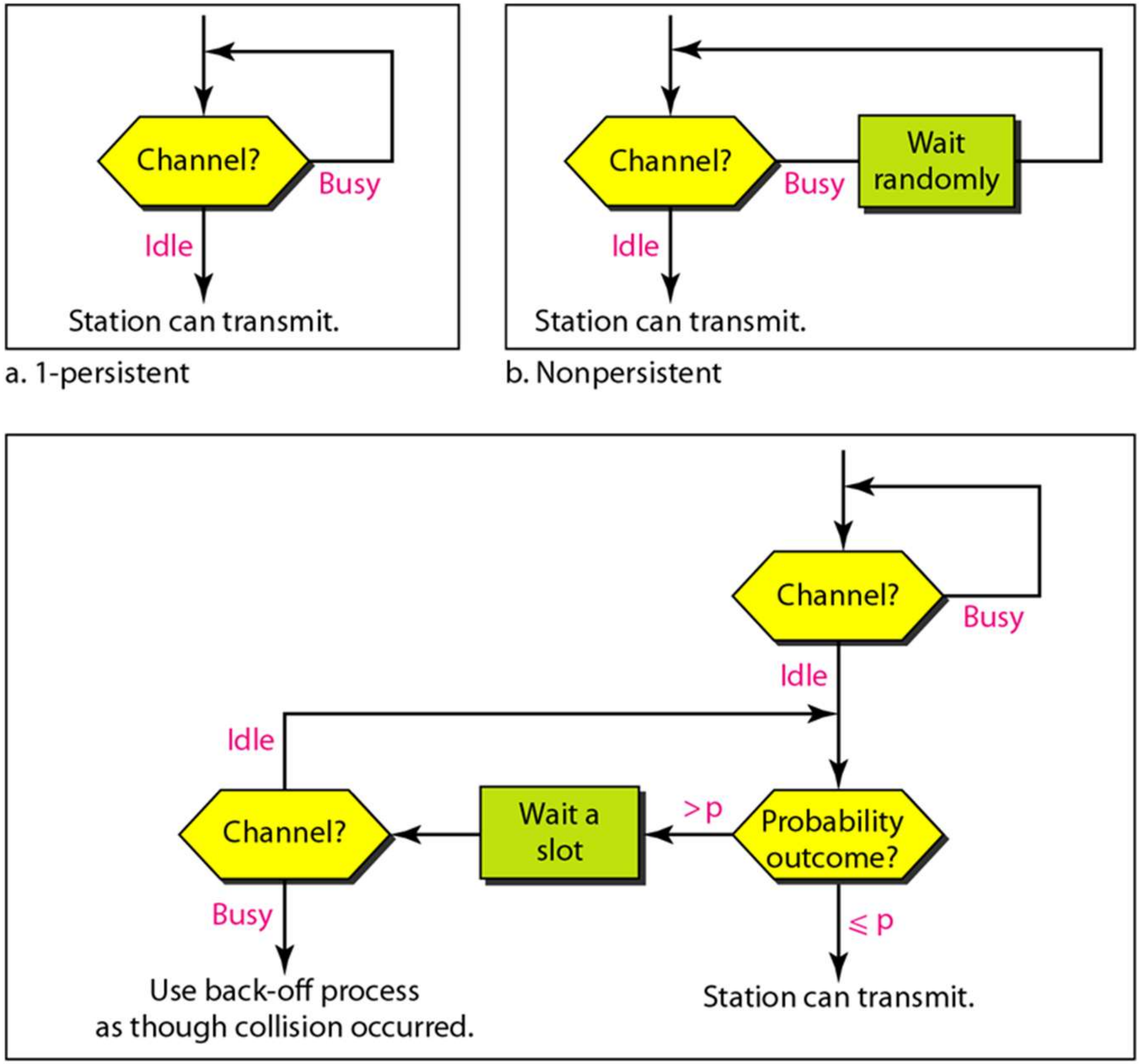
- Collision probability is low, but the node can waste time waiting even after the channel becomes idle

p-persistent CSMA

- While busy, the node continuously sense the channel until it becomes idle.
- If the channel becomes idle
  - The node transmits with probability p
  - The node defers transmission with probability 1-p



Three types of CSMA



c. p-persistent

Three types of CSMA

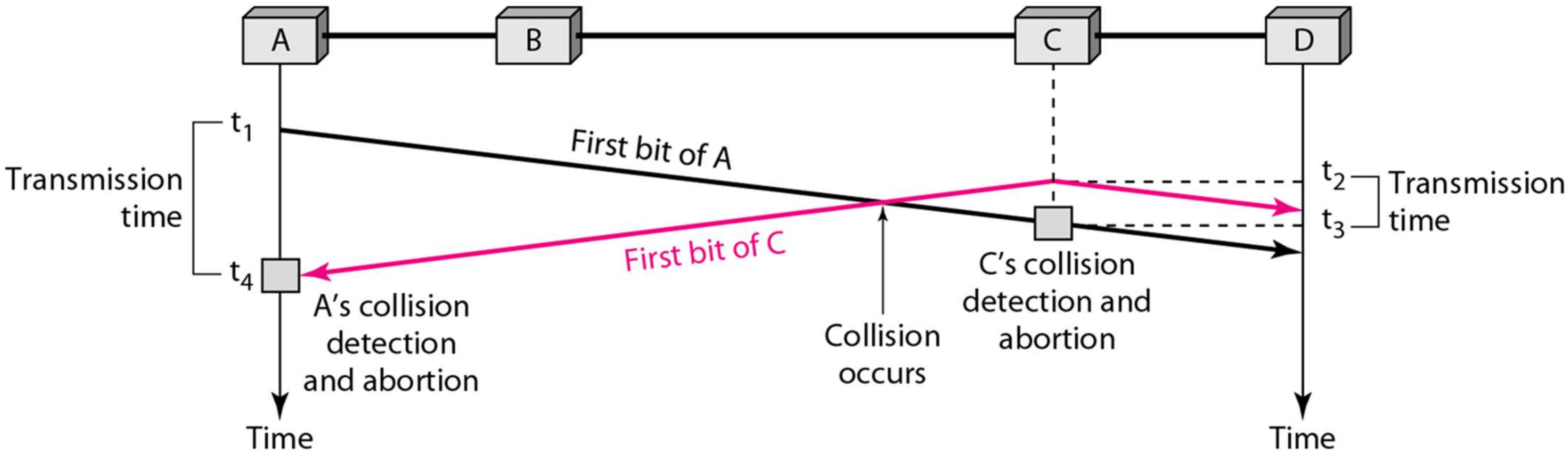
	1-persistent	Non-persistent	p-persistent
channel idle	transmit immediately	transmit immediately	transmit with prob. p
channel busy	keep sensing	wait random time, then sense again	keep sensing
collision probability	high	low	medium
channel utilization	high (aggressive)	low (conservative)	balanced

CSMA/CD (Collision Detection)

- Collision detection
  - If a collision occurs, the sender detects the collision and acts upon it
- Historically used in classic Ethernet (bus/hub topology)
- Modern switched Ethernet uses full-duplex, and thus does not use CSMA/CD

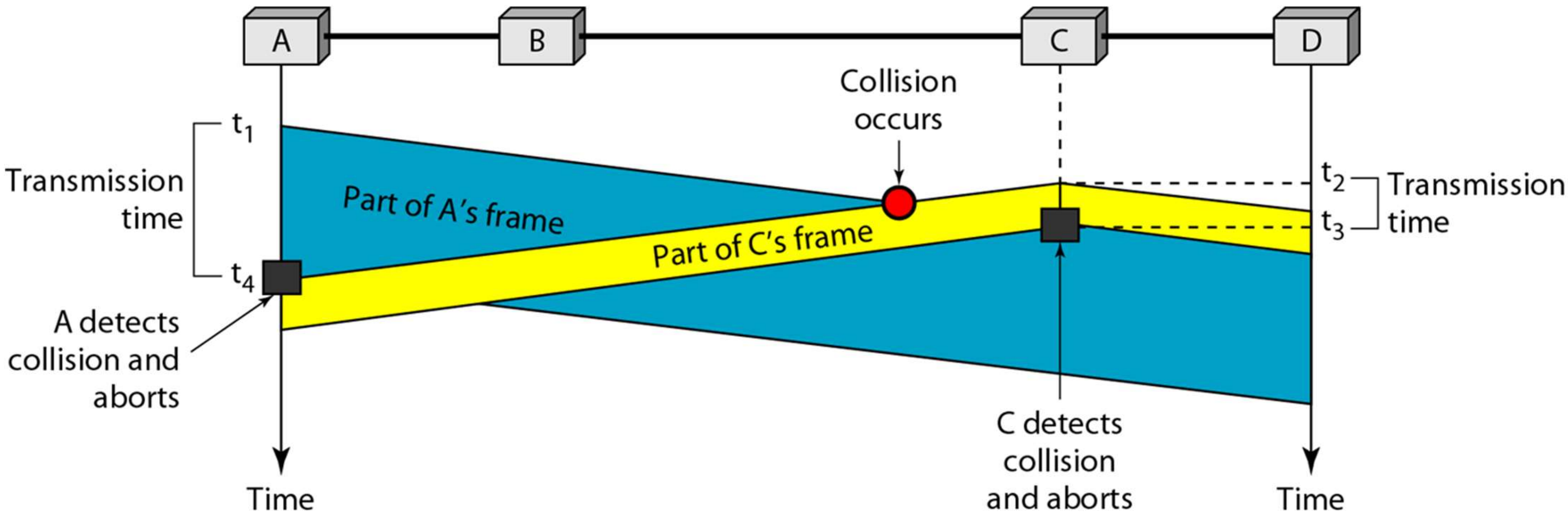
CSMA/CD

- Collision situation
  - Node A transmits a frame, but C transmits before the frame reaches C



CSMA/CD

- How a node detects collision
  - <u>While transmitting my frame, another frame arrives</u> → detect collision

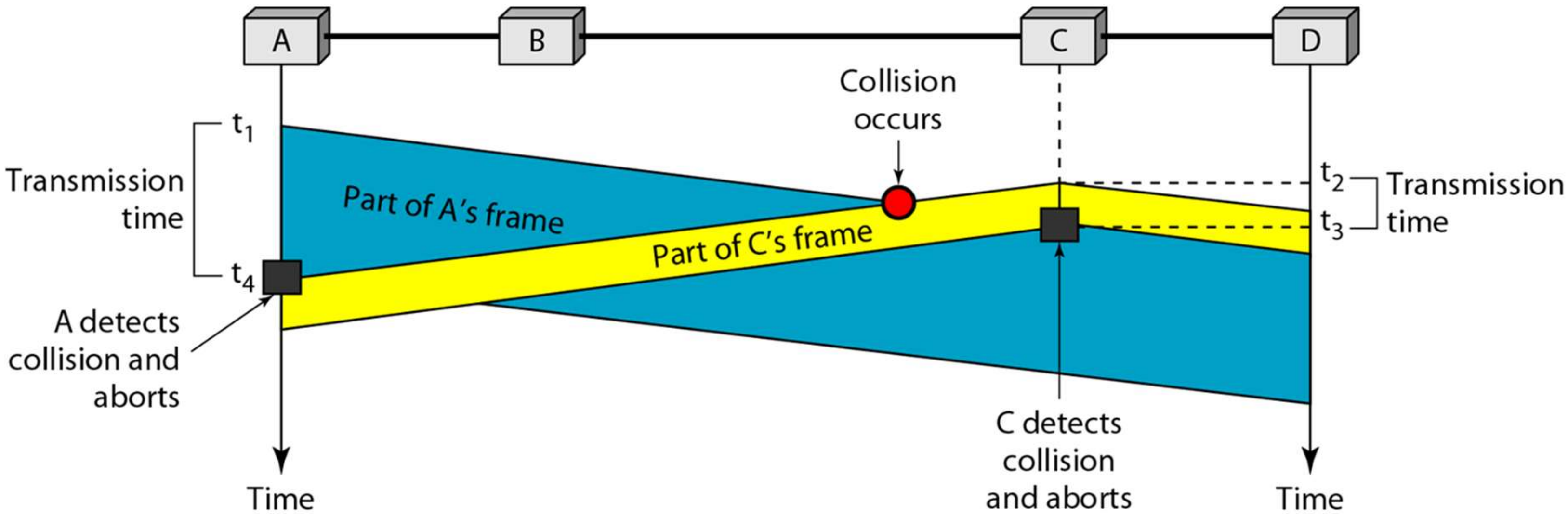


CSMA/CD

- Successful collision detection
  - If a collision occurs, the node should be able to detect it
  - <u>When the collided frame arrives, I should still be transmitting my frame</u>
  - If I finish my transmission too early, I might not be able to detect collision

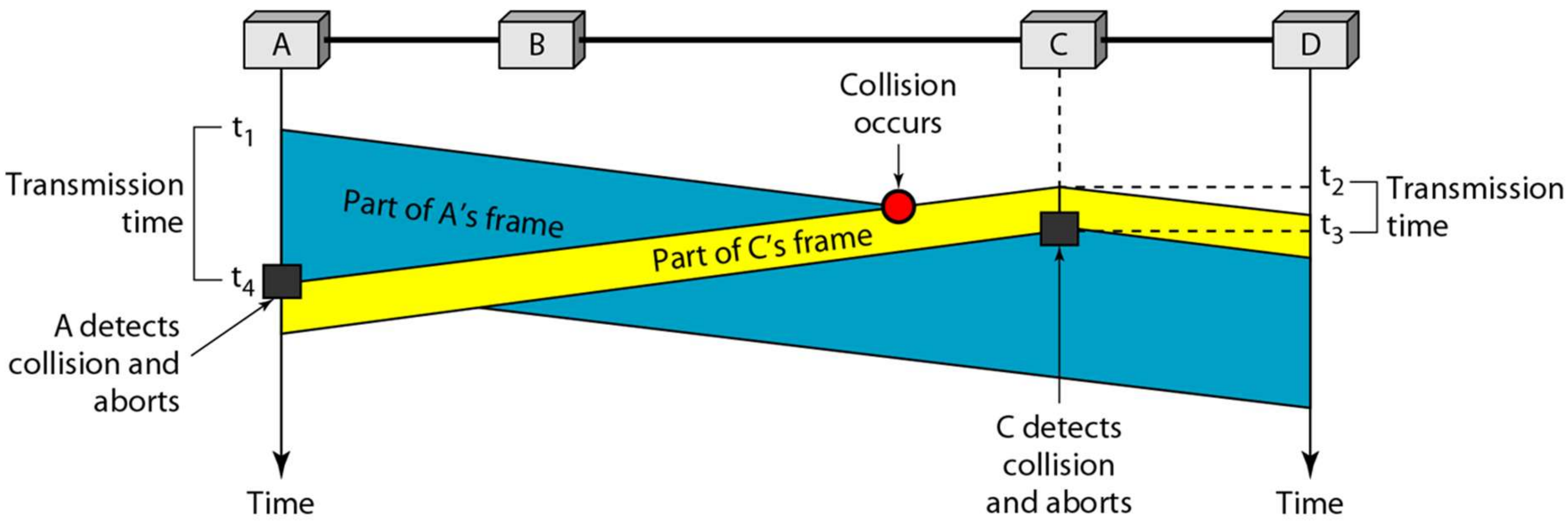
CSMA/CD

- Condition for collision detection failure
  - Node A transmits at t_1
  - Node C transmits at t_2 → collides with A's frame
  - The collided frame reaches node A at t_4
  - If node A finishes transmission before t_4, A fails to detect collision



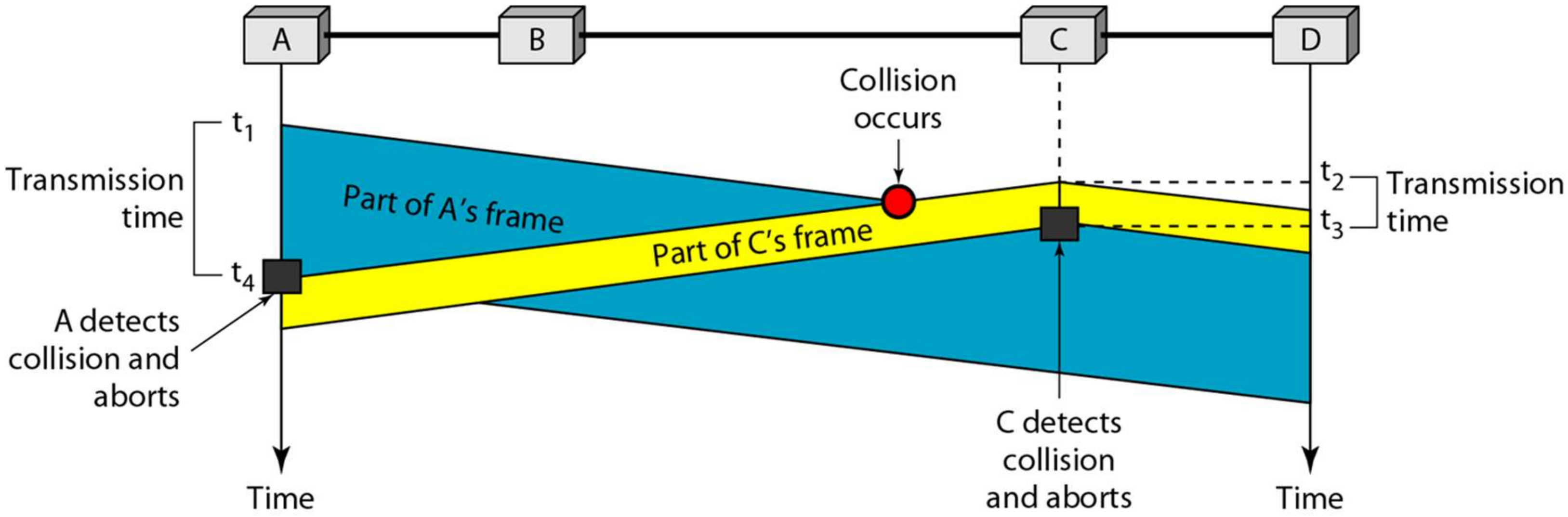
CSMA/CD

- Then how long should node A transmit?
  - If A starts transmission at t_1, the first bit reaches C at t_1+t_p
  - If C transmits before t_1+t_p, it is a collision
    - C cannot transmit after t_1+t_p because of carrier sensing.



CSMA/CD

- Then how long should node A transmit?
  - Suppose C transmits just before t_1+t_p.
  - The collided frame reaches node A at t_1+2t_p
  - At t_1+2t_p, node A should still be transmitting
  - Thus, $t_{fr} \geq 2t_p$

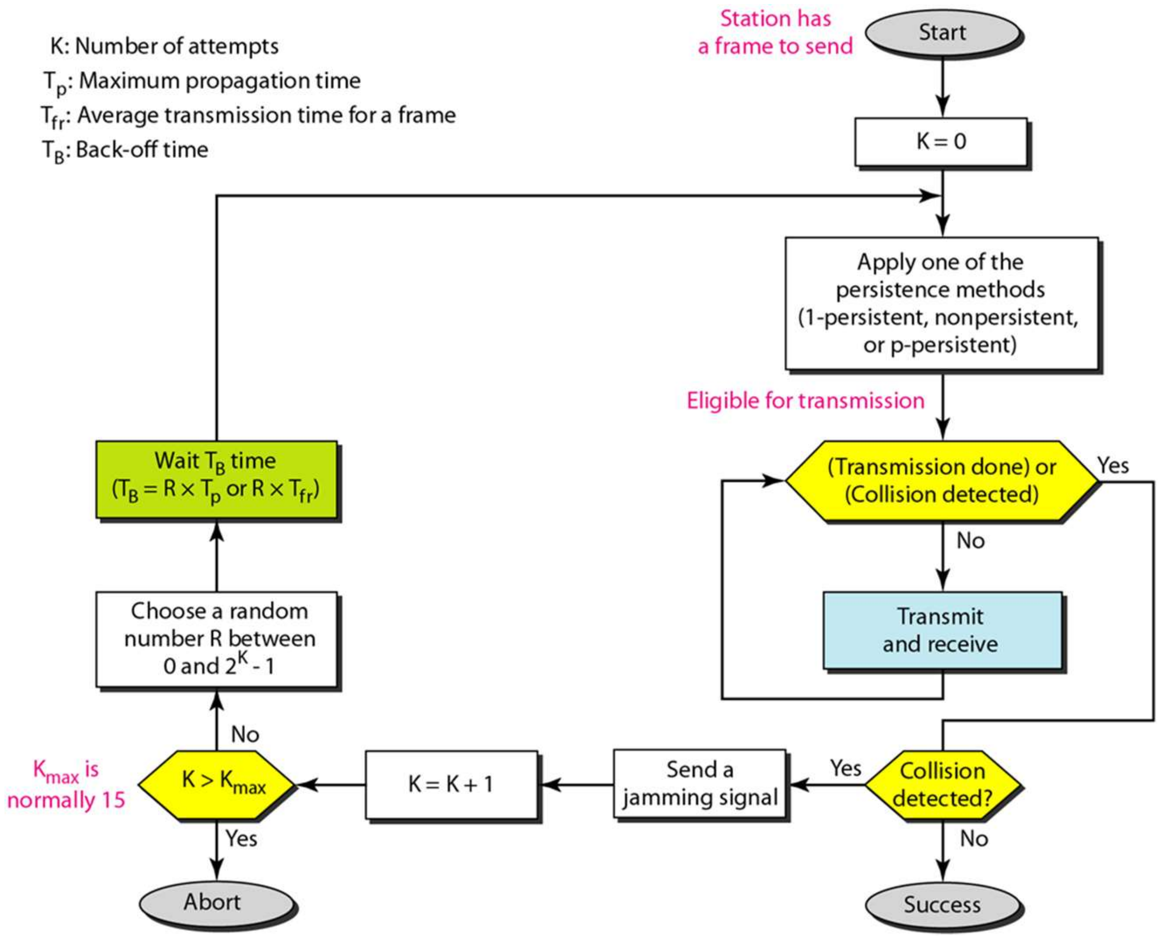


CSMA/CD

- A network using CSMA/CD has a bandwidth of 10Mbps. If the maximum propagation time is 25.6μs, what is the minimum size of the frame?

CSMA/CD

- When collision is detected
  - Send a jamming signal (notify all nodes)
  - Apply binary exponential backoff (similar to ALOHA)
  - Retry transmission



CSMA/CD

- When a node detects a collision, it sends a jamming signal
  - typically 48 bits of garbage data
- Purpose: Ensure all nodes know about the collision
  - Without jamming, other transmitting nodes might miss the collision and think the transmission succeeded
  - With jamming, the strong garbage signal propagates to all nodes
- All nodes then back off and retry with binary exponential backoff

CSMA/CA (Collision Avoidance)

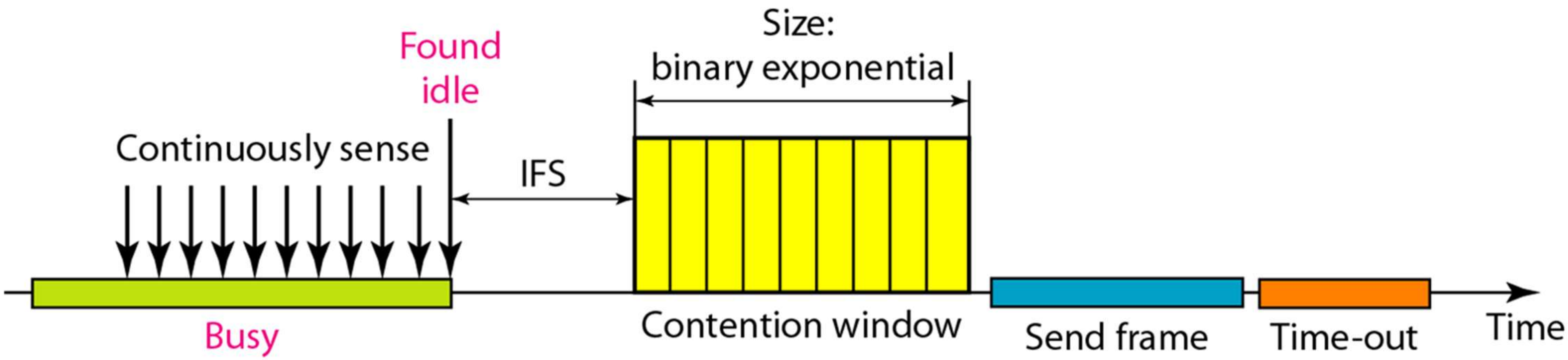
- Used in systems that cannot detect collision
- Used in Wi-Fi
- The Wi-Fi interface is half-duplex, which means it cannot transmit and receive at the same time → cannot detect collision as CSMA/CD

CSMA/CA

- Do random backoff before a collision occurs
- Wait for ACK
- If ACK does not come, regard it as collision → increase backoff exponentially

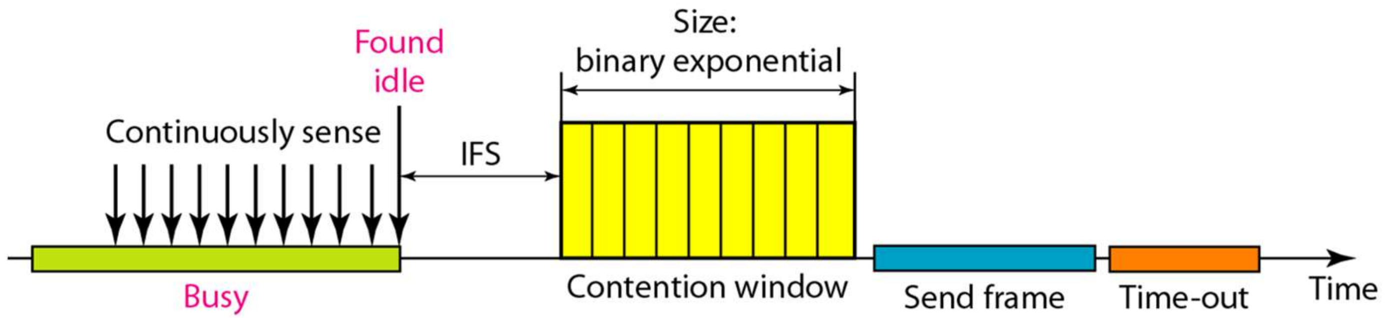
CSMA/CA

- IFS (inter-frame spacing)
  - Even if channel is sensed idle, another node might just started transmission. IFS is inserted in order to avoid collision.



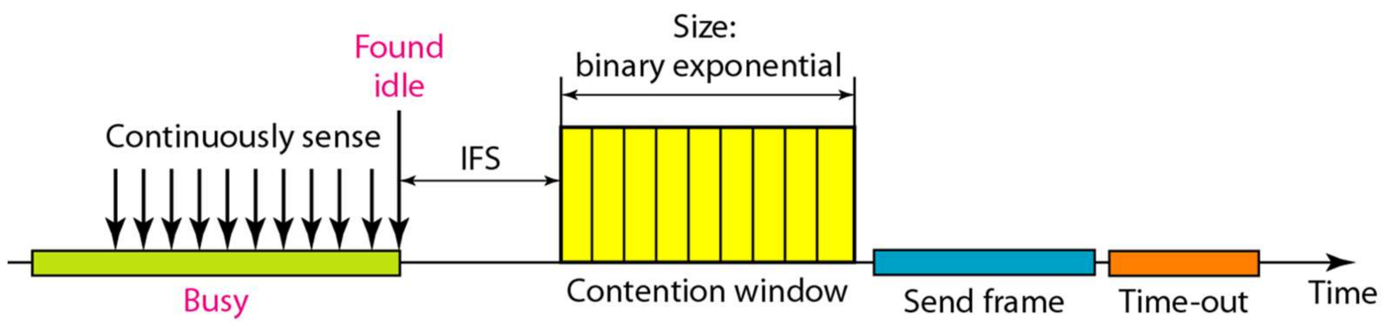
CSMA/CA

- Contention window
  - A range from where a node selects a random number (e.g. 0-31)
  - A node selects a number from the contention window
  - For each time slot, if the channel is idle, count down the number (e.g. time slot = 20us)
  - If the counter becomes zero, start transmitting.



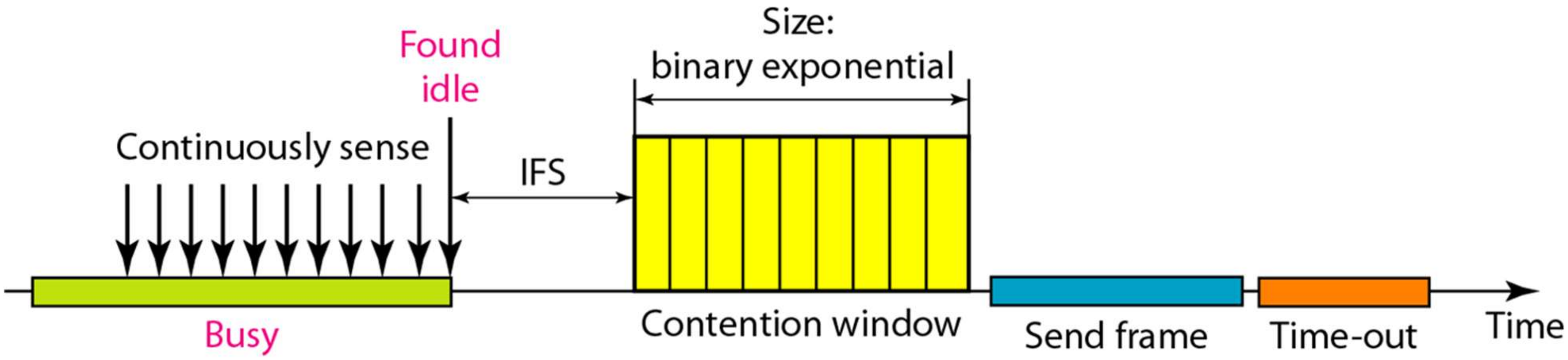
CSMA/CA

- Contention window
  - Suppose a node selects 15 as the random number (time slot = 20us)
  - If the channel is idle for 15x20=300us, the node transmits
  - If the channel becomes busy while counting, the node pauses until the channel becomes idle again
  - If the channel becomes idle, the node waits for IFS and then restart counting



CSMA/CA

- Binary exponential backoff
  - If a collision occurs, the contention window size doubles (e.g. 0-31 → 0-63)
  - If a frame is successfully transmitted, the contention window reduces to initial range



Example

- Assumptions
  - time slot = 20us
  - IFS = 50us
  - initial contention window: [0-31]
  - transmission time of a frame = 100us
  - propagation time = 0 (neglected)
  - no ACK
- Nodes A, B, C are in the same wireless LAN
  - If node A transmits, node B and C both can see the signal

- Assume the channel has just become idle (e.g., right after a previous transmission), and all three nodes have a frame waiting to send

Example

- 0us: A, B, and C has frames to send, they all wait for IFS (50us).

- 50us: A, B, and C selects a random number. A chooses 10, B chooses 15, and C chooses 20. They start counting down.

- 250us: A's counter becomes 0. A starts transmission. B and C see A's signal and <u>pause</u> count down.

Example

- 350us: A’s transmission finishes. B and C sense channel as idle. B and C wait for IFS.
- 400us: B and C resume counting. B’s counter = 5, C’s counter = 10
- 500us: B’s counter becomes 0. B starts transmission. C pauses counting.

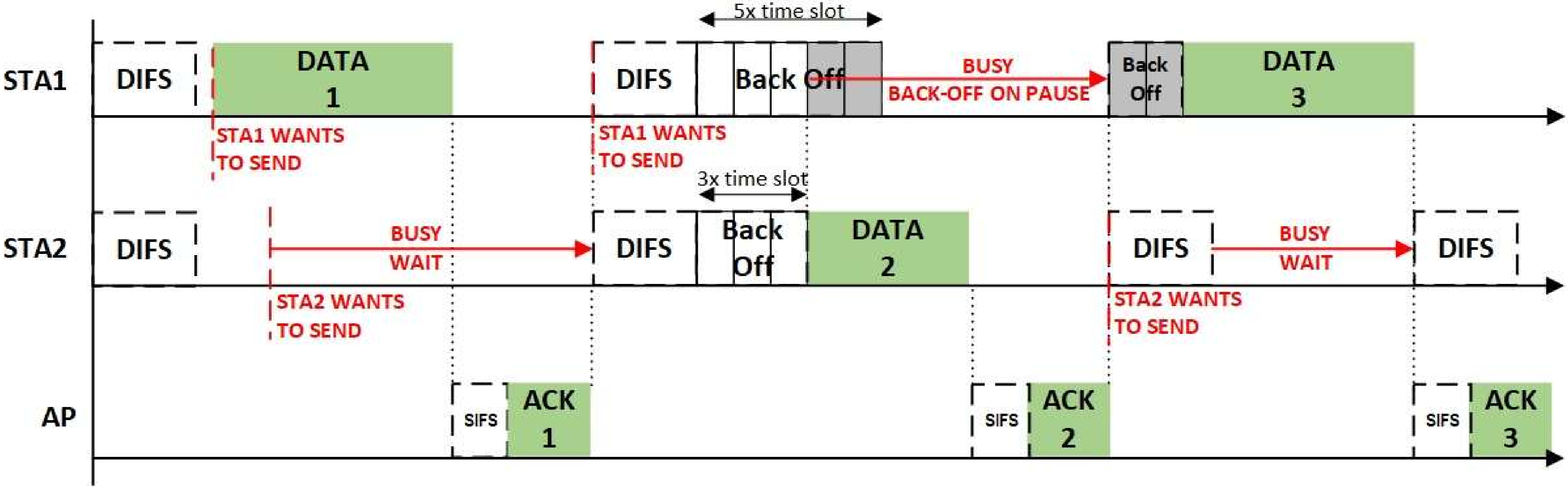
Example

- 600us: B’s transmission finishes. C waits for IFS.
- 650us: C resumes count down. C’s counter = 5
- 750us: C starts transmission
- 850us: C finishes transmission

CSMA/CA with ACK

- The previous example assumed no ACK for simplicity
- In real Wi-Fi, ACK is used to confirm successful reception
- SIFS (Short IFS): shorter than DIFS, used for high-priority transmissions like ACK
- Full sequence for a successful transmission:
  - Sender senses channel idle, waits DIFS + backoff
  - Sender transmits frame
  - Receiver waits SIFS after receiving frame
  - Receiver transmits ACK
  - Other nodes then wait DIFS to start their backoff
- Why SIFS shorter than DIFS?
  - To give ACK higher priority than new transmissions
  - Prevents other nodes from starting transmission before ACK completes

CSMA/CA with ACK



Controlled Access

Controlled Access

- Does not depend on random number to avoid collision

- Nodes agree on a rule to take turns and transmit

- Reservation
- Polling
- Token Passing

Reservation

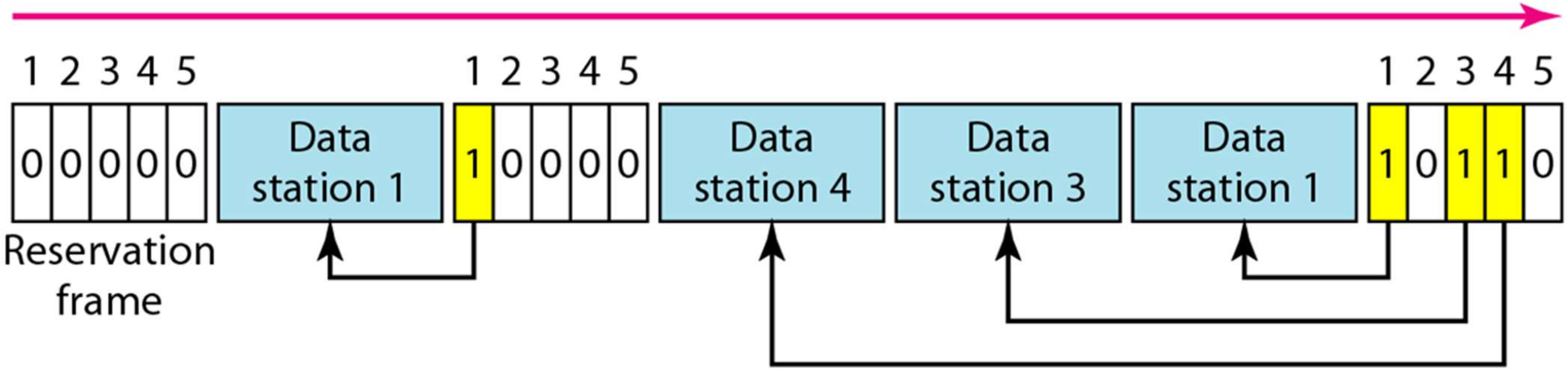
- Before transmission, a node "reserves" a time slot.

- There is a "reservation slot" before data slots, where nodes reserve their transmission.

- If there are N nodes, the reservation slot has N mini slots, where a node can send signal to announce that it will be transmitting.

Reservation

- In the first period, nodes 1, 3, and 4 reserve channel
- In the second period, only node 1 reserves channel

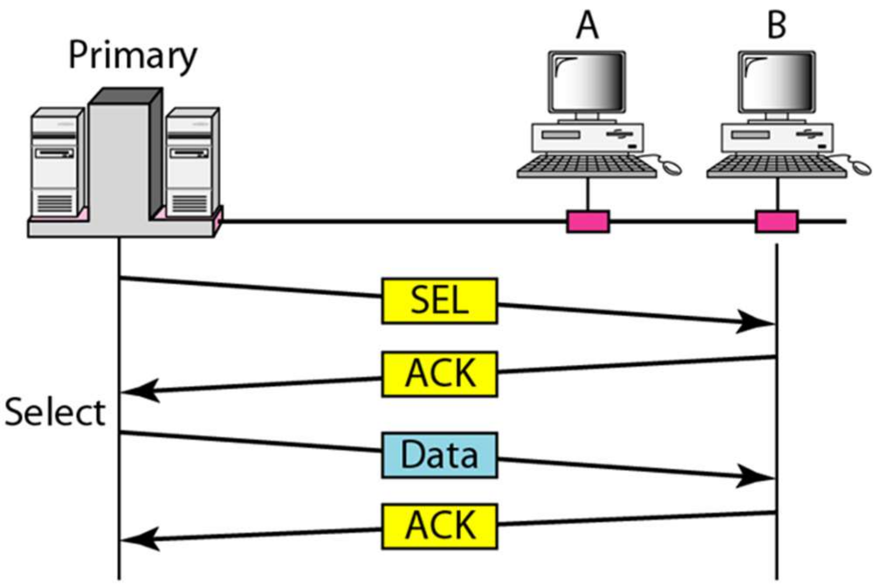


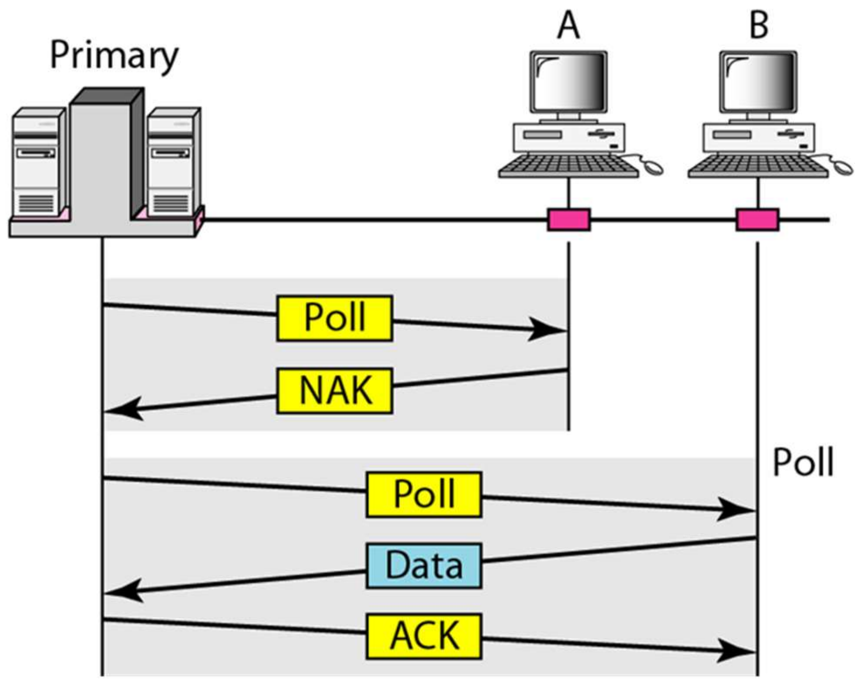
Polling

- One of the nodes become the “primary node”
- Other nodes become secondary nodes
- The primary node controls data transmissions
  – It decides who sends when
- The data should always go through the primary node
  – Two secondary nodes cannot communicate

Polling

- Select (SEL): a primary node sends data to a secondary node
- Poll: a secondary node sends data to a primary node



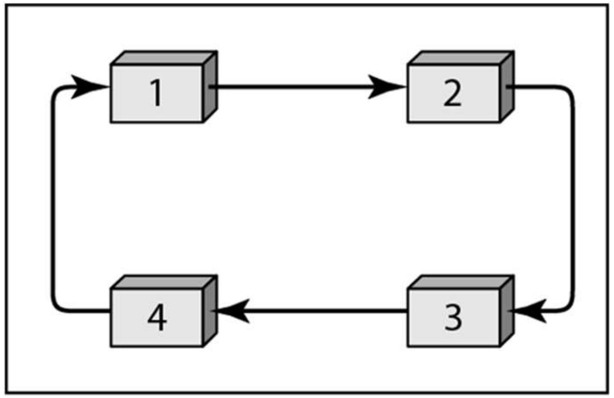


Token Passing

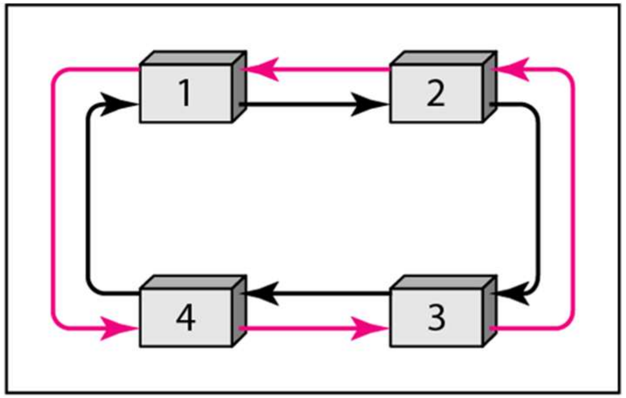
- The goal of medium access control is to have only a single node transmit at a time
- This mechanism uses a unique "token" to determine who should transmit.
  - There is only one token in the system
- A node can transmit only if it possesses the token. After data transmission, the node passes the token to the next node in the predefined sequence.

Token Passing

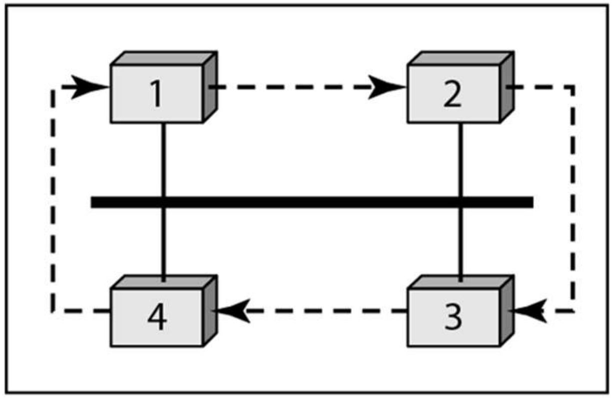
- Various types of token passing sequence (topology)



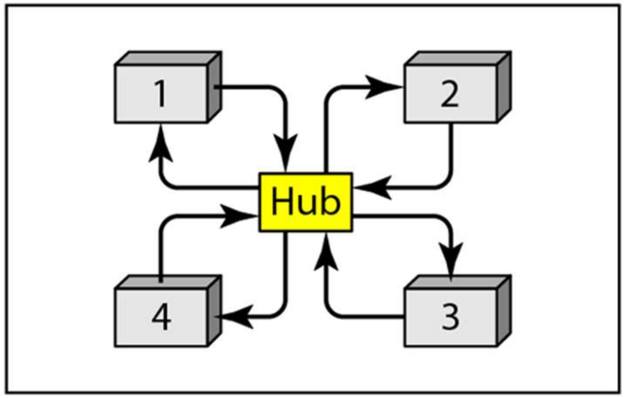
a. Physical ring



b. Dual ring



c. Bus ring



d. Star ring

Token Passing

- Physical ring
  - Simple, token does not need to have address of next node
  - If a link breaks, the system is stalled
- Dual ring
  - When a link breaks, the system can continue using the second link
- Bus
  - The token should indicate the next node
  - Tolerant to link break
- Star
  - The token is always passed through the hub
  - Easy to add/remove node from the system

Channelization

Channelization

- A mechanism to divide the medium into multiple channels using time, frequency, and code division techniques

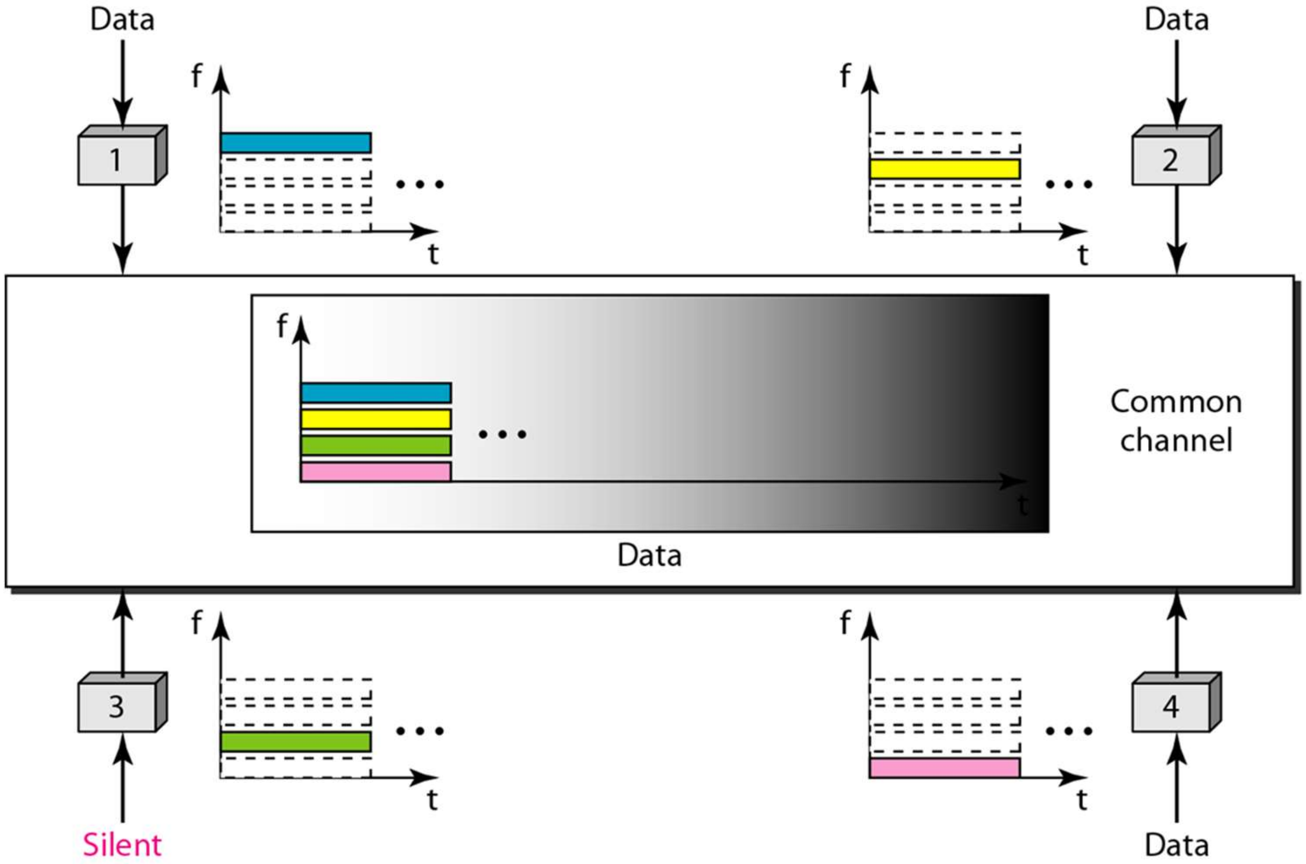
- Frequency-Division Multiple Access (FDMA)

- Time-Division Multiple Access (TDMA)

- Code-Division Multiple Access (CDMA)

FDMA

- Divide the bandwidth into multiple frequency sub-channels, and allocate sub-channels to different users



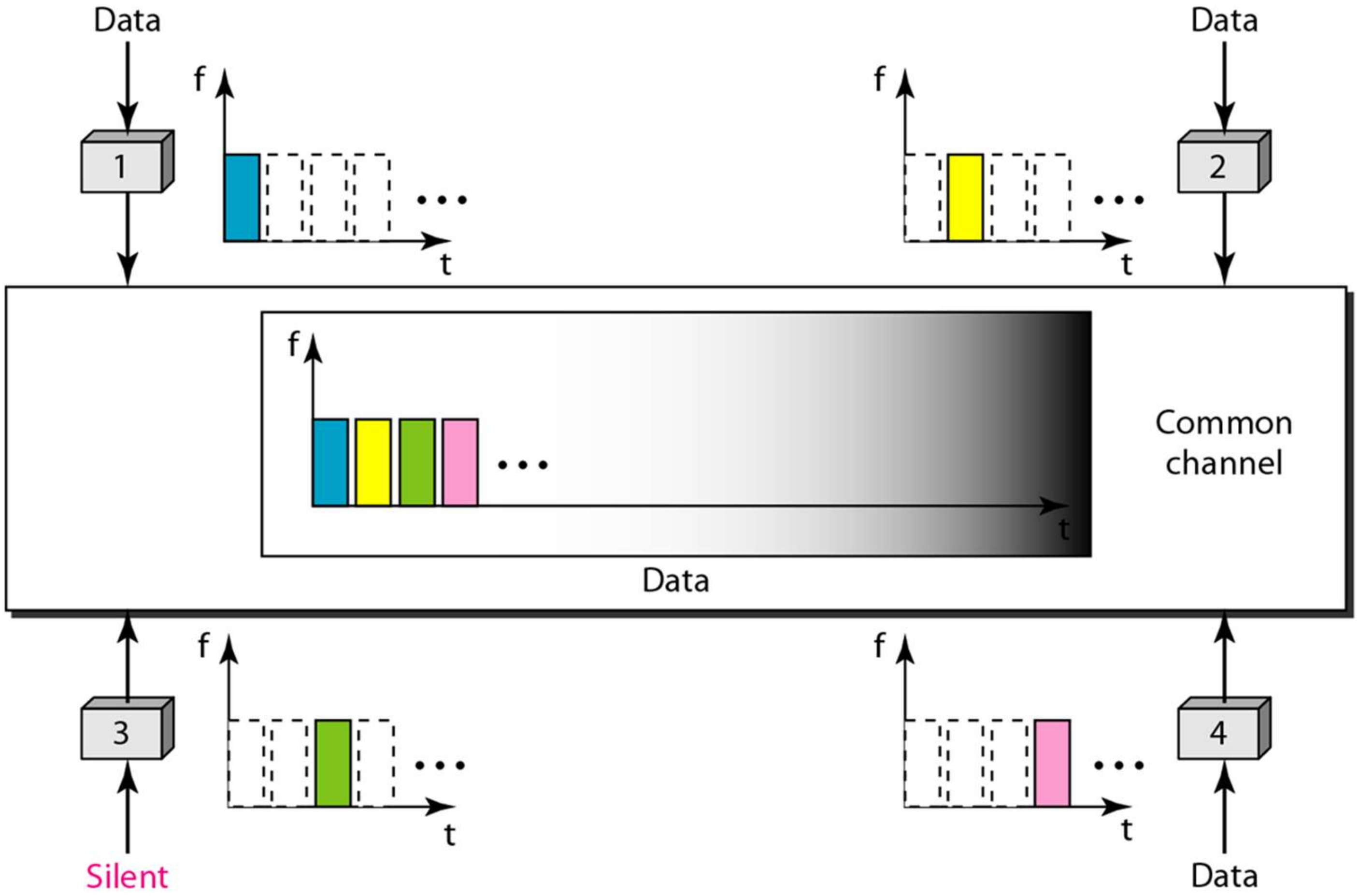
FDMA

- Guard band needed between sub-channels

- Suppose we divide a 20MHz channel into 1MHz sub-channels. If we need 100kHz of guard band, how many sub-channels can we make?

TDMA

- Divide the channel into time slots and allocate time slots to different users



TDMA

- TDMA is more flexible than FDMA, since we can change the allocation rate dynamically.

- However, TDMA requires time synchronization among nodes, which is a difficult problem.

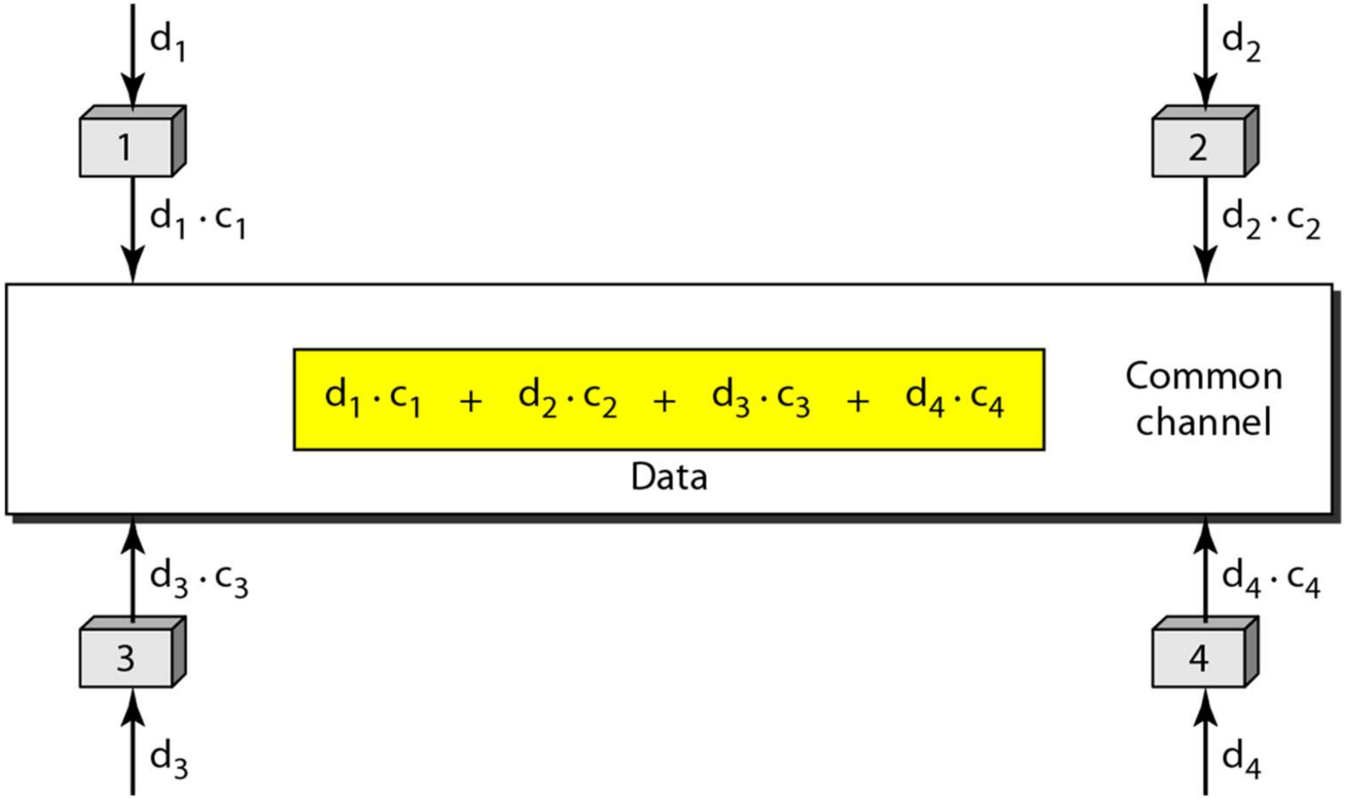
CDMA

- In CDMA, multiple nodes transmits at the same time, using the same frequency

- The signals will be mixed, but they can be separated using code division technique

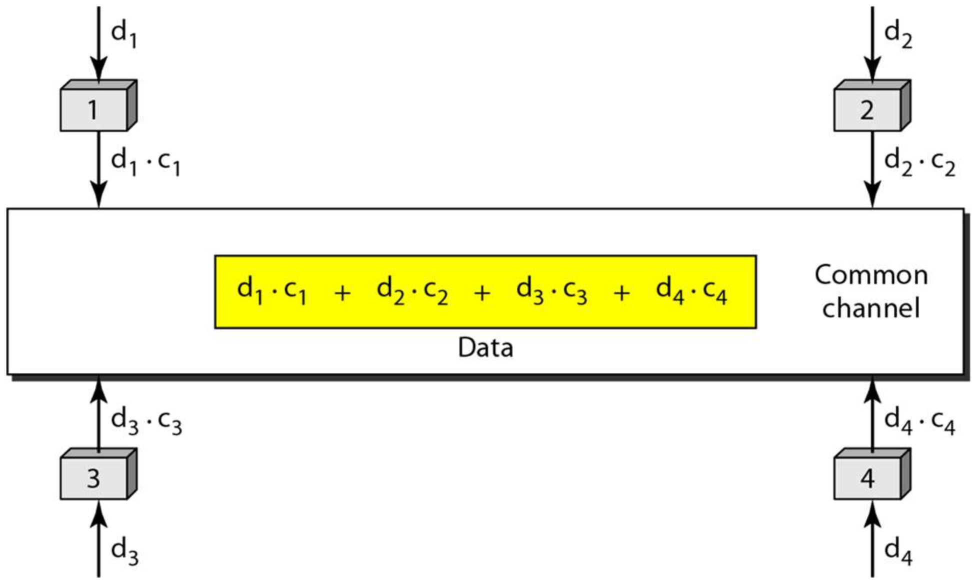
CDMA

- Suppose 4 nodes transmit data



CDMA

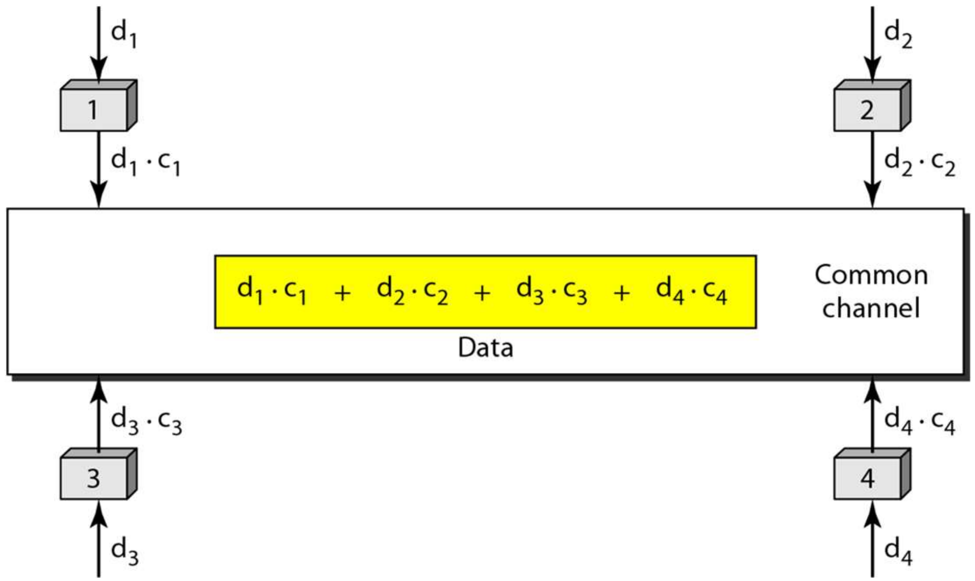
- Codes are allocated to nodes. Suppose c_i is the code allocated to node i. The codes have the following characteristics:
  - $c_i \times c_j = 0$, if $i \neq j$
  - $c_i \times c_i = 4$ (number of nodes)



CDMA

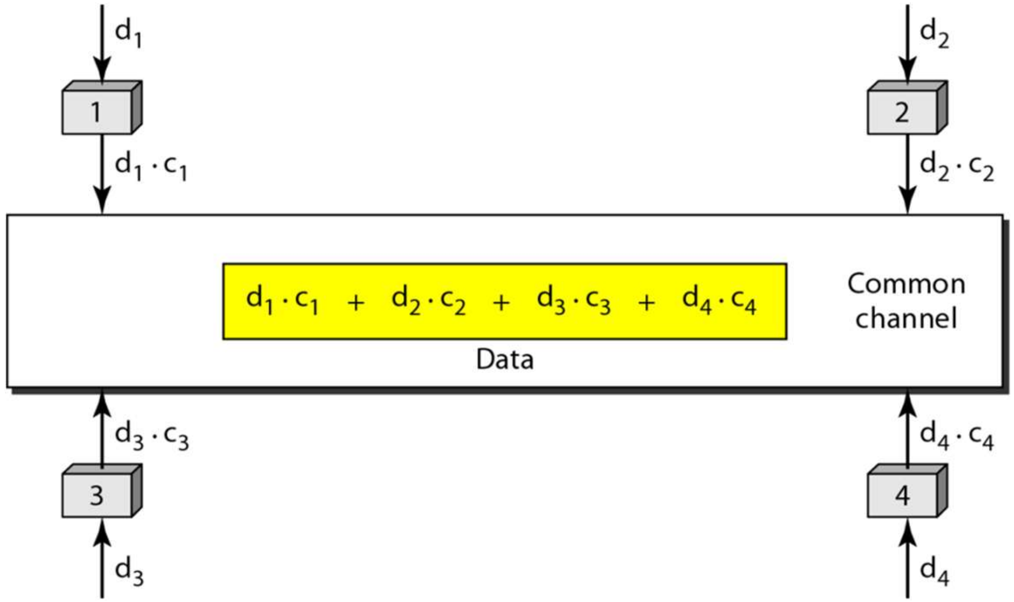
- When a node transmits data, it multiplies its data and its code. (d_i x c_i)
- If all four nodes transmit, the mixed data will be:

$$\boxed{d_1 \cdot c_1 + d_2 \cdot c_2 + d_3 \cdot c_3 + d_4 \cdot c_4}$$



CDMA

- The receiver multiplies the sender's code to the received data
- If a node wants to receive node 1's data, it multiplies c_1 and the received data
- Then, divide the result by 4 (number of nodes)



CDMA chips

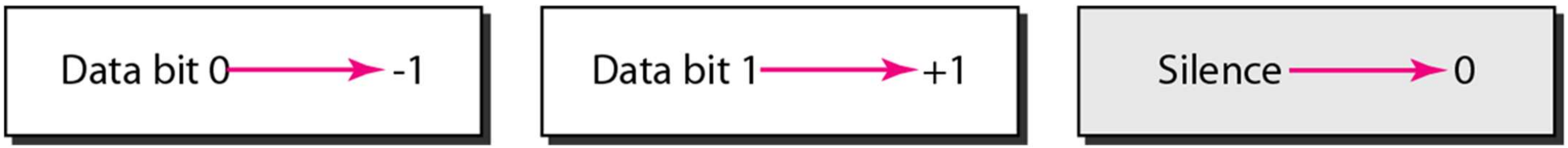
- The following code set can be used in CDMA

C_1	C_2	C_3	C_4
[+1 +1 +1 +1]	[+1 -1 +1 -1]	[+1 +1 -1 - 1]	[+1 -1 -1 +1]

- The code set has the required characteristics
  - If multiplied by itself (inner product), the result is 4.
  - If multiplied by other code, the result is 0.

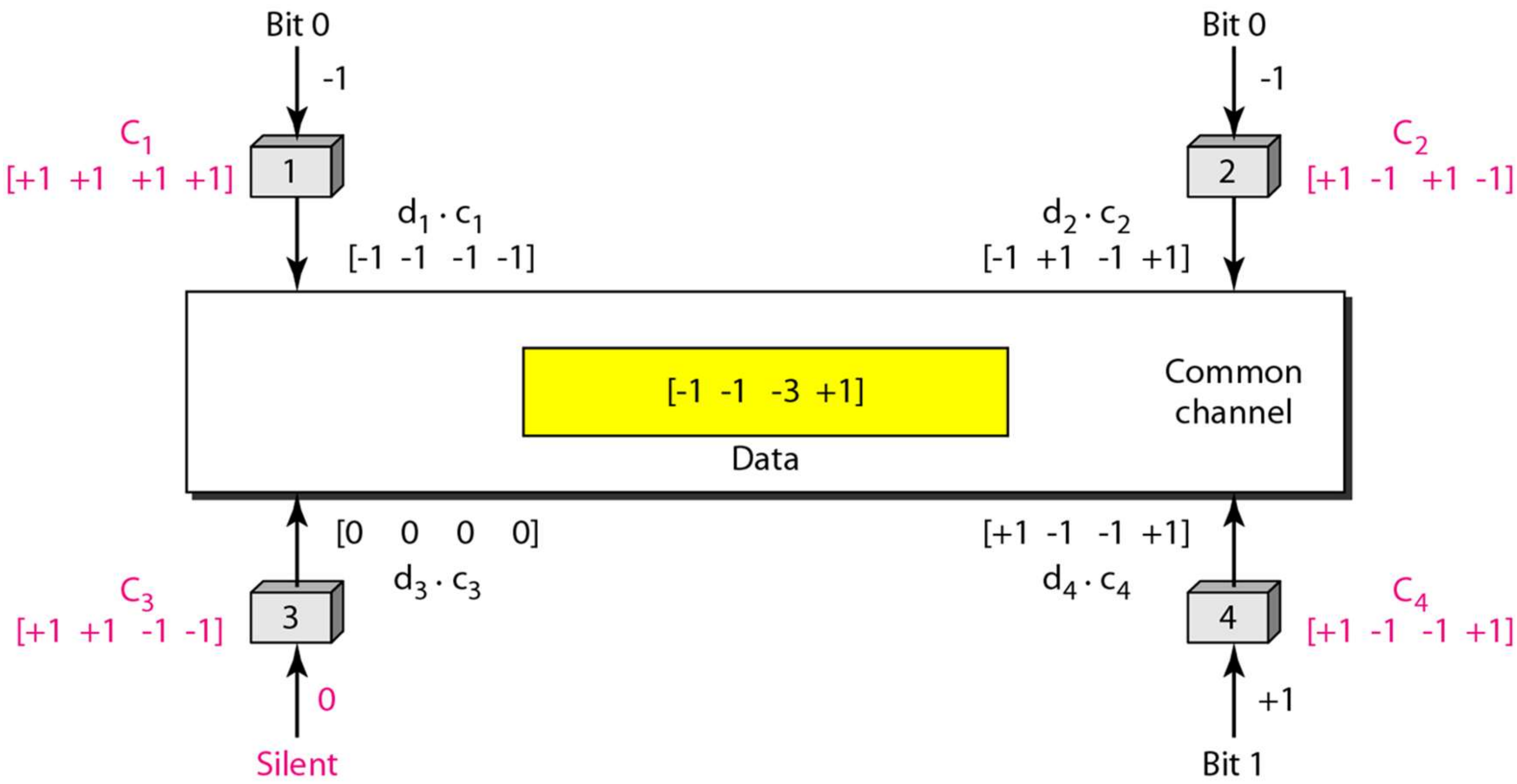
Transmission using CDMA

- To transmit data bits using CDMA, we make the following rules



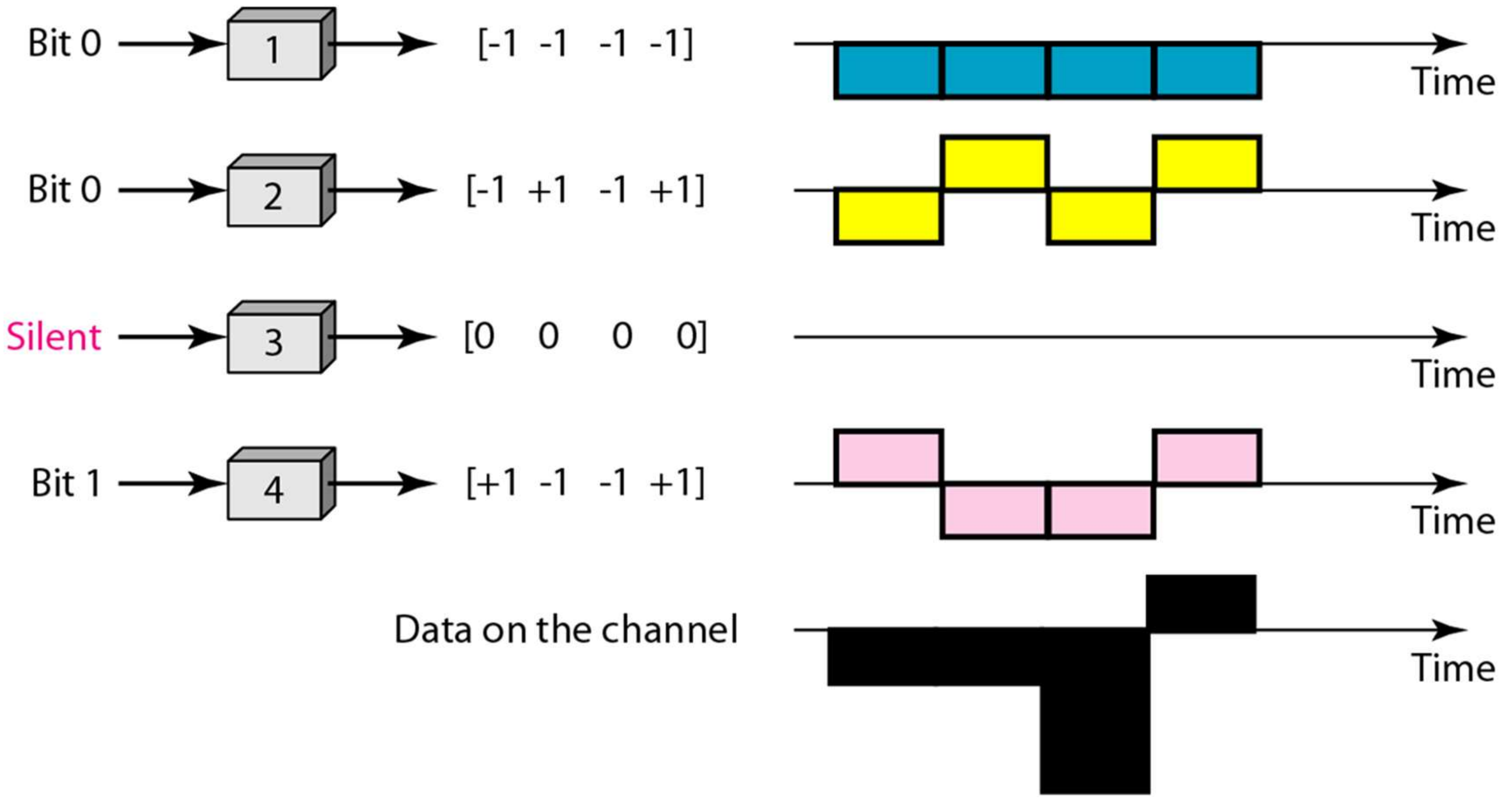
Transmission using CDMA

- At an instance, node 1 sends bit 0, node 2 sends bit 0, node 4 sends bit 1, and node 3 does not transmit.



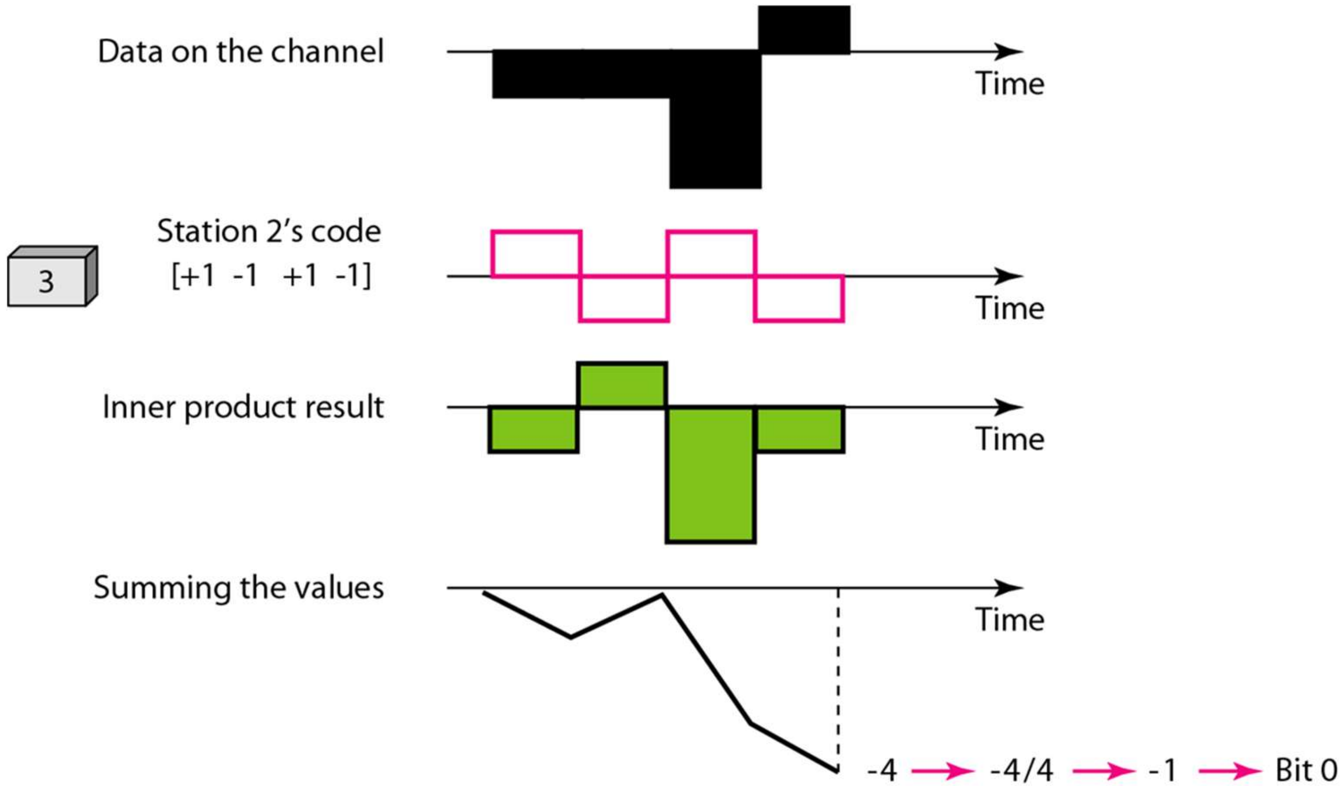
Transmission using CDMA

- The transmitted signals



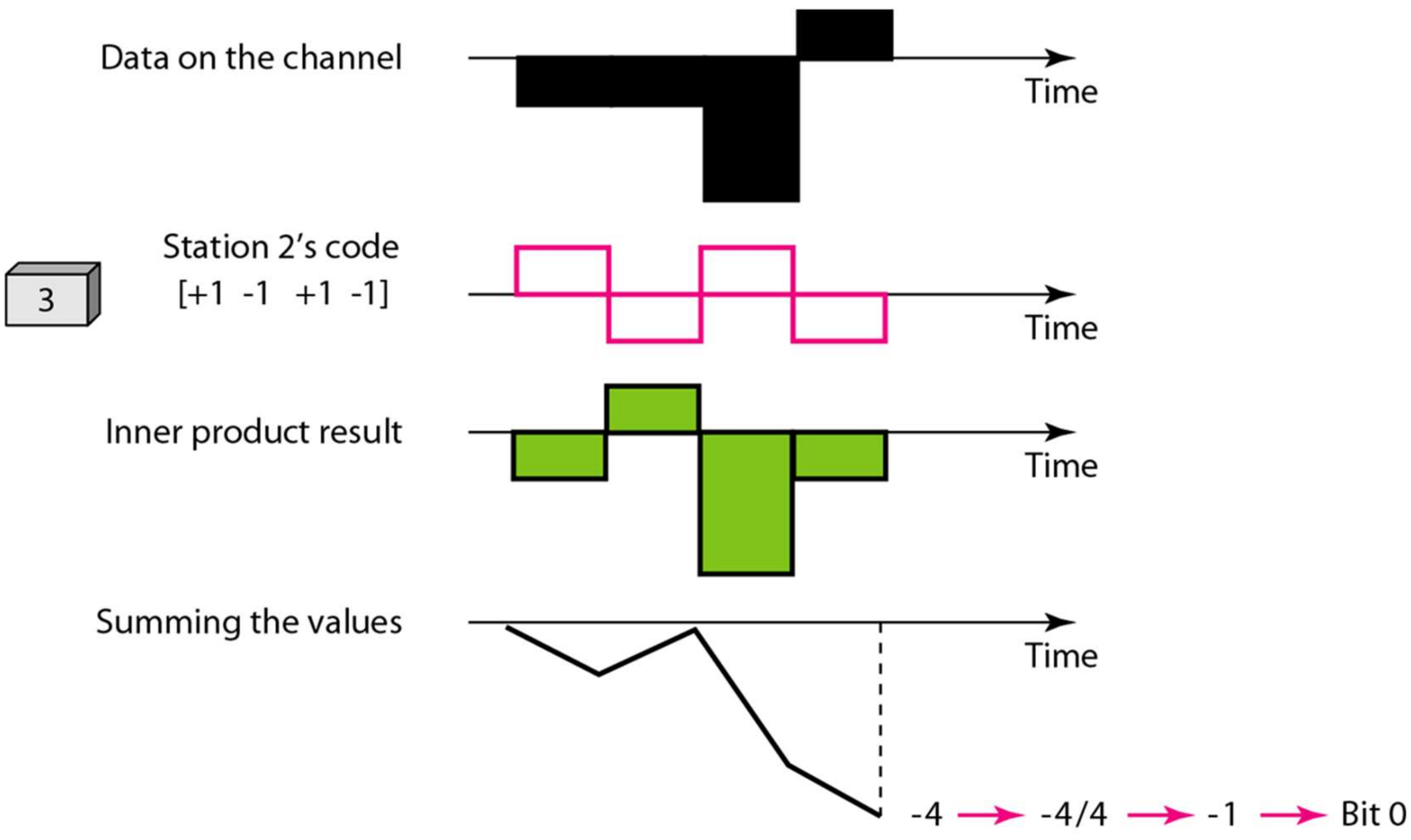
Transmission using CDMA

- Suppose node 3 wants to receive node 2's data
- Then, node 3 multiplies (inner product) received data with node 2's code



Transmission using CDMA

- Finally, add all numbers and divide by 4.
- The result is -1, which means bit 0.



Generating CDMA chips

- To generate CDMA chips, we can use Walsh table.
- To make Walsh table, we create a 2N x 2N matrix from N x N matrix, starting from a 1x1 matrix.
- Invert only the last (bottom-right) component in the generated matrix.

$$W_1 = \begin{bmatrix} +1 \end{bmatrix} \qquad W_{2N} = \begin{bmatrix} W_N & W_N \\ W_N & \overline{W_N} \end{bmatrix}$$

Generating CDMA chips

- Create 4x4 matrix from 2x2 matrix

$$W_1 = \begin{bmatrix} +1 \end{bmatrix}$$

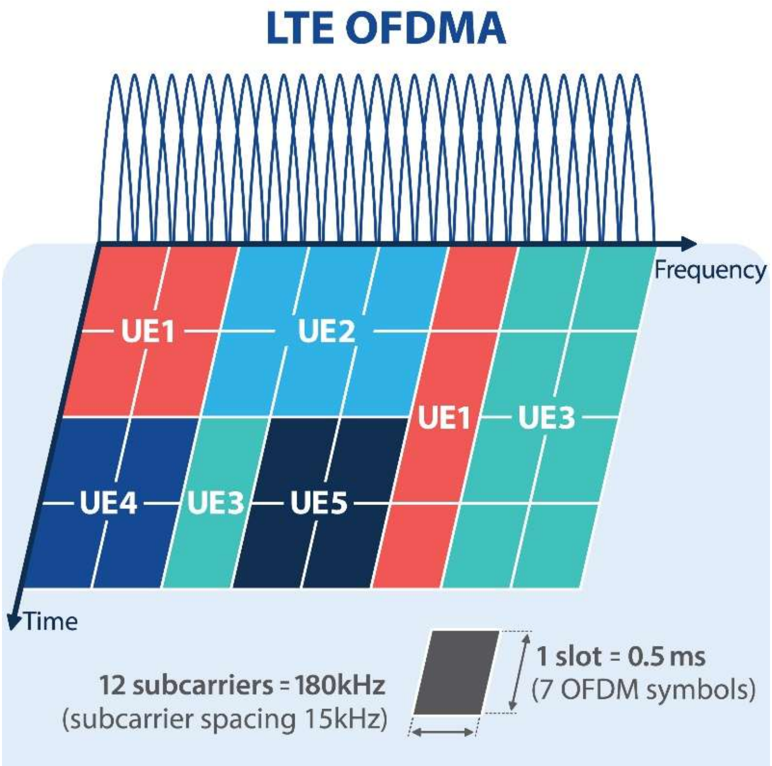
$$W_2 = \begin{bmatrix} +1 & +1 \\ +1 & -1 \end{bmatrix}$$

$$W_4 = \begin{bmatrix} +1 & +1 & +1 & +1 \\ +1 & -1 & +1 & -1 \\ +1 & +1 & -1 & -1 \\ +1 & -1 & -1 & +1 \end{bmatrix}$$

- The <u>columns</u> of the matrix become the CDMA codes.

Additional Topic: OFDMA

- OFMDA: Orthogonal Frequency-Division Multiple Access
- Based on OFDM (Orthogonal Frequency-Division Multiplexing)
- Divides the bandwidth into many narrow, orthogonal subcarriers
- Groups of subcarriers are allolcated to different users
- Can allocate resources in both frequency and time dimensions



OFDMA vs. FDMA

- FDMA
  - Each user gets a fixed frequency band
  - Requires guard band between bands to prevent interference
  - Wastes bandwidth on guard bands
- OFDMA
  - Subcarriers are orthogonal (mathematically non-interfering)
  - No guard band needed between subcarriers
  - Much more spectrum-efficient
  - Each user can be given a flexible number of subcarriers based on needs

OFDMA in Real Systems

- 4G LTE (2009): downlink uses OFDMA
- 5G NR (2019): both downlink and uplink use OFDMA
- Wi-Fi 6 (802.11ax, 2019): OFDMA introduced to Wi-Fi
  - CSMA/CA is still used to gain channel access
  - Once the AP has access, it can schedule multiple users to transmit simultaneously on different subcarrier groups
  - Much better in dense environments (many devices)

- OFDMA has become the standard channelization technique for modern wireless systems

End of Chapter